

Sun, Jun 21 at 8:21 AM

I noticed there is a bio-resonance device which threat the tuberculosis perfectly

Thought for 22s >

That claim is **not supported by reliable medical evidence**. Bioresonance devices have not been proven to diagnose, eliminate, or cure tuberculosis. Advertising authorities have found no robust clinical evidence that these devices can treat diseases, and health agencies warn against replacing proven treatment with unverified therapies.  ASA +1

Tuberculosis is caused by *Mycobacterium tuberculosis*. Active TB requires a combination of specific antibiotics —typically a **4-, 6-, or 9-month regimen**, selected according to drug-susceptibility testing and the patient's condition.  CDC +1

A device reading cannot confirm that TB is gone. Proper evaluation uses clinical assessment, chest imaging, sputum molecular testing, culture, and drug-resistance testing.  CDC +1

Please do not stop or delay prescribed TB medication because of a bioresonance treatment. That could allow the disease to worsen, become drug-resistant, or spread to other people. The words “**treats tuberculosis perfectly**” are a serious red flag. Share the device's name, website, or brochure and I'll examine exactly what evidence its manufacturer provides.

    ...   Sources

can you find a product catalog for biomolecular nuclear magnetic resonance 21th edition

Thought for 34s >

I found the likely catalog:

Cambridge Isotope Laboratories — “Stable Isotopes for Biomolecular NMR”

It is a 64-page product catalog covering isotope-labeled media, amino acids, protein standards, RNA/DNA reagents, cell-free expression products, and deuterated buffers.  Chemie Brunsch...

The manufacturer's official catalogs-and-product-listings page is also available here.  isotope.com

I could not verify a catalog titled exactly “**Biomolecular Nuclear Magnetic Resonance 21st Edition**.” You may mean the **2021 Biomolecular NMR catalog**, or a differently named commercial device. Also, the catalog above is for **scientific laboratory reagents**, not a bioresonance machine that treats tuberculosis.

A photo of the device label, manufacturer name, or its exact model number would let me locate the correct catalog.

    ...  Sources

I saw a mobile device which measure human health. the device was named with biomolecular nuclear magnetic resonance 21th edition.

Stopped thinking >



I saw a mobile device which measure human health. the device was named with biomolecular nuclear magnetic resonance 21th edition

Thought for 31s >

Ah—now I know the device you mean. It is most likely the **21G+ Quantum Resonance Magnetic Body Analyzer**, sometimes advertised with the grammatically incorrect phrase “21th generation.” One matching product is the **NIVKART 21G+ Quantum 2-in-1 Resonance Magnetic Body Analyzer**. Its listing says the user places a palm on the unit or holds a sensor rod, after which software produces 54 health reports. [nivkart.com](#)

This is **not genuine biomolecular nuclear magnetic resonance equipment**. Scientific biomolecular NMR uses large, high-field laboratory spectrometers to study molecular structures; it is not a small hand-held health scanner. [Division of Rese...](#)

Also, terms such as “21G,” “21st generation,” “quantum,” and “magnetic resonance” are marketing names, not proof that the device uses clinical MRI/NMR technology. The advertised “up to 90% accuracy” appears to be a seller claim, not a result established through independent clinical trials. [nivkart.com](#) Regulators reviewing similar bioresonance devices report that they have not seen robust evidence showing that these machines can diagnose or treat diseases. [ASA](#)

So this device should **not** be used to diagnose tuberculosis, confirm that TB has disappeared, or replace blood, sputum, imaging, or laboratory testing. TB is a bacterial infection that requires proper testing and antibiotic treatment. [World Health Or... +1](#)

The closest product catalog page I found is the **NIVKART 21G+ product listing** cited above. Because these units are commonly rebranded by different sellers, a photo of the device, box, software screen, or manufacturer label would help identify the exact catalog and manual.



21G+ Quantum Resonance Magnetic Body Analyzer, this is just I wanted to find

Thought for a few seconds >

Great—that pins it down. The device is commonly listed as:

21G+ Quantum Resonance Magnetic Body Analyzer

Also called **21st/21th Generation Quantum Magnetic Resonance Analyzer** or **Quantum Health Analyzer**.

These devices are often sold under different brand names, so the manufacturer label or software version matters when finding the correct manual or product catalog. It is not clinical NMR/MRI equipment and should not be relied on to diagnose or treat tuberculosis or other diseases.

Shall I find the exact **21G+ catalog, software manual, and specifications**?



yes, please

Thought for 32s >

I found the closest matching **catalog, user manual, and specifications** for the device.

Product catalog

NIVKART 21G+ Quantum 2-in-1 Resonance Magnetic Body Analyzer

The catalog page identifies it with SKU/ASIN **B0C6QSLV8G** and says it uses either the palm plate or handheld testing rod. It advertises a one-year warranty and 52–54 generated reports. Oddly, the same page uses both numbers, suggesting that the software/report package may vary by seller. At the time checked, it was listed at ₹8,999 and marked sold out. 

User manual PDF

I found an **18-page “Quantum Resonant Magnetic Analyzer” user manual** that appears to cover the software family supplied with these devices. It includes:

- Software installation
- Windows 7, 8 and 10 instructions
- USB encryption-key/dongle setup
- Creating customer records
- Starting a test and holding the sensor rod
- Viewing, printing and exporting reports
- Comparing two test reports
- Database backup and restoration
- Sensitivity adjustment
- Cleaning and maintenance

The manual is available through this PDF result:  Amazon Media

The basic workflow described is:

Connect analyzer → insert USB key → launch software as administrator → create client → hold sensor rod → start test → save and generate report

Listed hardware specifications

The matching NIVKART listing provides these details:

Item	Listed specification
Model/part number	21G+
Manufacturer/brand	NIVKART
Product type	2-in-1 analyzer
Connection	Normally USB
Computer requirement	Windows PC
Dimensions	Approximately 35 × 9.5 × 27 cm
Weight	Approximately 1.65 kg
Country of origin	China
Report count	Advertised as 52 or 54
Accessories	Sensor rod/palm sensor, USB cable, software and USB security key
Warranty	One year

The dimensions, weight, part number and country-of-origin information come from the associated product listing. 

I did **not** find a manufacturer-issued PDF explicitly titled “21G+ 21st Edition Manual.” These machines appear to be generic hardware rebranded by different sellers, and the included software version may differ. The safest way to get the exact software is from the USB drive or disc supplied with the particular unit; random downloads may not recognize its security dongle.

One important warning: the advertised 85–90% accuracy and disease-detection claims are seller claims, not established clinical accuracy. Regulators say they have not seen robust evidence that bioresonance devices can diagnose or treat medical conditions. It should not be used to diagnose tuberculosis or determine whether TB has been cured.  [asa.org.uk](#) +1

    ...   Sources

The Quantum Resonance Magnetic Analyzer (QRMA) is a non-invasive diagnostic device that uses quantum physics principles and electromagnetic signals to analyze the body’s health. The device measures the resonance of various frequencies in the body, with the idea that different tissues and organs emit distinct magnetic fields. By comparing these frequencies to standard healthy ranges, the QRMA provides insights into various health conditions.

The analyzer claims to detect imbalances in vital organs, nutrient levels, and body functions, offering a holistic view of the user’s health. It is often marketed for its potential to provide quick, comprehensive health

Show more 

Thought for 9s >

That paragraph describes **how the QRMA is marketed**, but it presents the claims as established facts. A more accurate description is:

The Quantum Resonance Magnetic Analyzer is a wellness device marketed as using a handheld sensor and software to generate reports about organs, nutrients, and general health. Sellers claim that it detects electromagnetic or “quantum resonance” patterns and compares them with reference values. However, these mechanisms and diagnostic claims have not been validated by reliable clinical evidence. The device should not be regarded as medical NMR, MRI, or a clinically proven diagnostic instrument.

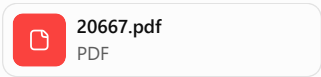
There is no accepted medical evidence that individual organs emit unique magnetic frequencies that this small device can measure through the hand and use to diagnose dozens of conditions. Advertising regulators have repeatedly found bioresonance diagnostic and treatment claims unsupported.  [ASA](#) +1

The FDA has also warned manufacturers of comparable hand-sensor systems that software outputs identifying diseases amount to medical diagnoses and therefore require proper medical-device authorization and supporting evidence.  U.S. Food and D...

So these phrases should be treated carefully:

Marketing statement	Evidence-based interpretation
“Uses quantum physics”	A marketing phrase; not proof of a scientifically validated quantum measurement
“Measures organ frequencies”	Not an established clinical diagnostic method
“Detects nutrient deficiencies”	Cannot reliably replace blood or laboratory tests
“Provides a comprehensive diagnosis”	Produces software-generated wellness reports, not a confirmed medical diagnosis

Marketing statement	Evidence-based interpretation
"Non-invasive"	Likely true in the ordinary sense that it does not pierce the skin, but that does not prove accuracy
For tuberculosis specifically, QRMA results cannot establish whether someone has TB or has been cured. WHO-recommended diagnosis relies on clinical evaluation and tests such as rapid molecular testing of sputum or other appropriate samples, with imaging and additional laboratory testing where needed. who.int +1 DA Sources	



Thought for 12s >

I reviewed the paper. **It does not provide credible clinical validation of the 21G+ QRMA device.**

20667

The paper describes a prototype combining a commercially available QRMA unit, Raspberry Pi, Firebase, an Android app, and voice-to-text AI. Its main engineering contribution is data collection and presentation—not proof that the QRMA can accurately diagnose disease.

Several red flags stand out:

- The authors say the device is “validated” and “certified for medical use,” but provide no certificate number, regulator, testing laboratory, or supporting study.
- There is no comparison against blood tests, imaging, sputum testing, physician diagnosis, or another accepted clinical reference standard.
- No patient sample size, inclusion criteria, randomization, sensitivity, specificity, false-positive rate, or false-negative rate is reported.
- The references are mostly documentation for Kotlin, Firebase, TensorFlow, Raspberry Pi, and deep-learning books—not clinical research supporting QRMA.
- The claimed **85–95% accuracy** is unclear and appears to concern AI/audio analysis, not the QRMA sensor itself.
- More seriously, the table in **Figure 17 on page 10** shows reported model accuracy rising only to about **0.164, or 16.4%**, which conflicts sharply with the paper’s 85–95% claims.
- The paper itself admits that QMRA reliability is disputed and that further validation is required.

So the paper may demonstrate a software prototype that stores and displays QRMA-generated reports, but it does **not** prove that the underlying measurements are medically accurate.

For tuberculosis, this paper provides no evidence that QRMA can diagnose TB, measure treatment response, or confirm cure. TB still requires recognized clinical testing such as molecular sputum tests, culture, imaging, and physician assessment.

Sources

Mon, Jun 22 at 7:53 AM

invest all the magazines related to the Quantum Resonance magnetic analysis.
I need to find the database or AI model to measure the health metrics with the resonance frequency

Thought for 1m 57s >

I investigated the accessible journals, papers, manuals, and databases around **Quantum Resonance Magnetic Analysis/Analyzer, QRMA/QMRA**.

Bottom line

I could not find a public, scientifically validated database or AI model that maps the handheld 21G+ device’s alleged “resonance frequencies” to its 50-plus health metrics.

The most relevant direct validation study compared QRMA blood-glucose output with capillary glucose testing and found **no significant correlation**. Its conclusion was that QRMA did not represent the participants’ actual blood-glucose condition.  Mendeley +1

That does not prove every possible frequency-based health measurement is impossible. It means the **21G+ QRMA ecosystem is not presently supported by the kind of raw-data documentation, reference database, calibration studies, and independent clinical validation needed for medical use.**

Journal and paper investigation

Publication	What it studied	Evidence value
Jurnal Ners, 2019/2020 — QRMA versus anamnesis and capillary glucose	44 participants; directly compared QRMA glucose coefficients with conventional glucose measurements	Most informative validation study found. QRMA results did not significantly correlate with glucose measurements.  Mendeley +1
International Journal of Environmental Research and Public Health, 2021	Used QRMA outputs to compare active and less-active university students	It used QRMA as a measurement tool but did not validate it against laboratory tests. Its QRMA reference was a commercial equipment listing rather than an instrument-validation study.  MDPI
Journal of Clinical Rheumatology Research, 2022	Compared QRMA zinc results with colorimetric serum testing in 60 children	Reported 84% sensitivity and 87% specificity, but this is one small study and does not validate other QRMA reports, devices, generations, diseases, or nutrient measurements. Independent replication is needed.  ResearchGate +2
IJFMR, 2024, uploaded paper	Combined a commercial QRMA unit with Raspberry Pi, Firebase, voice recording and an Android app	Primarily an engineering prototype. It did not provide a proper QRMA-versus-clinical-reference validation. Page 2 acknowledges reliability concerns; pages 3–4 claim validation/certification without documenting the regulator or certificate; page 10 claims 85–95% while its displayed model table reaches only about 16.4% accuracy.  20667
Biomedicines, 2025	Used QRMA-generated insulin, cholesterol, triglyceride, HDL and LDL values in cardiovascular-risk research	The study did not compare the QRMA lipid outputs with laboratory blood assays. It repeats the manufacturer-style “85% accuracy” claim rather than demonstrating it.  MDPI
Publiciana, 2018	Descriptive screening of women’s reproductive health using QRMA	No accepted reference test was used to verify claims such as ovarian cysts, cervical cancer or vaginitis; therefore it cannot establish diagnostic accuracy.  journal.unita.ac.id
QRMA in nutritional practice, 2024	Reports experience with 50 weight-management clients	Descriptive use, not a controlled validation against DXA, blood testing or another reference method.  ResearchGate

The pattern is important: most favorable papers **use the QRMA-generated values as data**. They do not first prove that those values correspond to actual glucose, cholesterol, vitamin concentrations, organ function or disease.

The 2019 Jurnal Ners paper is different because it tested one QRMA output against a real reference measurement—and the QRMA failed that comparison.

What database is probably inside the 21G+ software?

The product manuals describe a handheld sensor, a short scan and software-generated coefficients compared against preset ranges. However, I found no public documentation specifying:

- raw sensor waveform or spectrum;
- actual excitation frequencies;
- magnetic-field strength;
- ADC resolution and sampling rate;
- measurement units;
- calibration standards;
- equations connecting signals to organs;
- training population;
- ground-truth laboratory tests;
- model weights or source code.

The vendor’s “database” is therefore probably a **proprietary lookup table of report ranges and text interpretations**, rather than a published biomedical spectral database. That is an inference from the available manuals and publications; the manufacturer has not disclosed enough technical material to verify it. The manual itself describes the mechanism using radio-tuning analogies rather than a reproducible measurement protocol. 

Real databases that use genuine resonance or frequency-dependent signals

These are scientifically useful, but they are **not plug-compatible with the 21G+ QRMA output**.

1. NMR metabolomics databases

These are the strongest match to genuine magnetic-resonance health analysis.

Database	Content	Suitable use
Human Metabolome Database, HMDB	Human metabolites, disease associations, experimental NMR spectra, peak lists and downloadable FID files	Identifying and quantifying metabolites from real serum, urine or other biofluid NMR spectra.  hmdb.ca +1
MetaboLights	Public metabolomics studies, reference spectra, biological roles and downloadable experimental datasets	Training disease-specific or biomarker-specific models using NMR or mass-spectrometry data. 
Biological Magnetic Resonance Bank, BMRB	NMR spectral and quantitative data for biological molecules and metabolites	Molecular spectral identification, chemical shifts and biomolecular research.  bmr.io +1

These require a real NMR spectrometer or properly acquired NMR spectra from biological samples. A palm sensor cannot use these databases unless it genuinely generates comparable calibrated NMR spectra—which the 21G+ has not been shown to do.

2. Bioimpedance spectroscopy databases

This is a more plausible direction for compact handheld hardware.

Bioimpedance spectroscopy measures complex electrical impedance across a known set of frequencies. Frequency-dependent resistance and reactance can provide information related to fluid distribution, cell membranes and body composition, under controlled conditions. It does not automatically reveal glucose, tuberculosis, vitamins or the health of every organ.  [PMC](#) +1

A public Zenodo compilation contains bioimpedance spectroscopy measurements from different instruments and includes plant, phantom and human-tissue measurements. It is useful for signal-analysis development, although it is not a ready-made clinical health-label database. 

3. Electrical impedance tomography and spectroscopy

EIT/EIS is used for narrowly defined tasks such as lung ventilation monitoring or tissue characterization. Open phantom datasets and some small human research datasets exist, but models are normally designed for one application—not a universal whole-body diagnosis.  [arXiv](#) +1

AI models and software you can actually use

Genuine NMR route

A credible NMR pipeline could use:

- **nmrglue** to read FID/spectral files, perform Fourier transformation, phase correction, baseline correction and peak processing;  nmrglue.readthe... +1
- **HMDB/BMRB** for reference peak matching;
- **Bayesil-style spectral profiling** for metabolite identification and concentration estimation;
- **MetaboAnalyst** for PCA, PLS-DA, random forest, SVM, statistical testing and biomarker analysis.  metaboanalyst.ca +1

This can support endpoints such as metabolite concentration or disease-risk classification when trained and validated against clinical diagnoses.

Bioimpedance route

For frequency-sweep impedance data, begin with the **Cole/Cole–Cole physical model**, not a deep neural network. Extract features such as:

- low- and high-frequency resistance;
- reactance peak;
- characteristic frequency;
- phase angle;
- dispersion parameter;
- electrode-contact stability;
- temperature and hydration covariates.

Then use interpretable models first:



Deep learning becomes reasonable only after acquiring a large, independent dataset with raw spectra and reference labels.

What you would need to build your own validated model

Do not train against the QRMA report values themselves. That would merely teach the AI to imitate unverified software.

Build one endpoint at a time:

Target	Required reference
Body fat or lean mass	DXA or validated multi-compartment assessment
Total/extracellular body water	Dilution method or validated clinical BIS equipment
Blood glucose	Laboratory plasma glucose or calibrated glucose analyzer
Cholesterol and triglycerides	Certified laboratory lipid panel
Vitamin/mineral concentration	Appropriate serum laboratory assay
Lung ventilation	Spirometry/EIT reference data

Target	Required reference
Tuberculosis	Molecular test, culture and clinical assessment—not impedance alone

For a compact “resonance-frequency” device, the most realistic starting technology is **multi-frequency bioimpedance spectroscopy**, not nuclear magnetic resonance. Genuine NMR requires specialized magnetic-field and radio-frequency hardware; a small handheld contact sensor is far more likely to measure electrical impedance.

1. Write the Target Measurement Specification

Create a one-page document like this:

Project:
Multi-Frequency Bioimpedance Research Device V0.1

Intended purpose:
Measure the body's electrical impedance spectrum through skin electrodes.

First health metric:
Example: total-body-water or extracellular-water estimate.

Target users:
Adults aged 18–65.

Body measurement location:
Hand-to-foot, hand-to-hand, or localized four-electrode measurement.

Raw measurements:
Frequency
Resistance
Reactance
Phase angle
Signal quality
Electrode contact impedance
Temperature

Reference measurement:
A recognized clinical or laboratory method.

Output:
Research estimate only—not diagnosis or treatment advice.

Acceptance target:
Repeatability, accuracy, bias, and failure-rate limits to be defined before collecting data.

Do **not** begin with “50 health reports.” Select one endpoint.

Best first endpoints

Candidate metric	Practicality	Suggested reference
Body water/hydration trend	Good	Validated clinical BIS or dilution reference
Body composition	Moderate	DXA or validated body-composition system
Local edema/fluid change	Good research target	Clinical assessment plus imaging/reference measurement
Respiration through impedance	Possible	Respiratory belt or spirometry
Blood glucose	Poor initial target	Laboratory plasma glucose
Vitamin levels	Poor initial target	Laboratory blood assay
Tuberculosis	Not suitable for impedance-only detection	Molecular test, culture and imaging

I would start with **hydration or fluid-distribution measurement**.

2. Decide whether it is wellness or medical

Write the exact claim before designing hardware.

A wellness claim might be:

"Displays impedance and hydration trends for general wellness research."

A medical claim might be:

"Detects dehydration" or "diagnoses a disease."

In the United States, intended use and indications for use determine whether a product is a medical device. Products claiming to diagnose, mitigate, treat or prevent disease fall outside ordinary general-wellness positioning.  U.S. Food and D... +1

That decision affects your development cost, clinical evidence, quality system and regulatory pathway.

3. Prove the sensor physics before testing people

Build a **bench measurement prototype** first.

You will need:

```

Programmable frequency generator
↓
Safe current-limited excitation circuit
↓
Four-electrode measurement interface
↓
Instrumentation amplifier
↓
Voltage/current or I/Q measurement
↓
ADC
↓
Impedance calculation
↓
USB/Bluetooth raw-data export
  
```



Prefer a **four-electrode, tetrapolar arrangement**:

- Two electrodes inject the test signal.
- Two separate electrodes measure voltage.
- This reduces the effect of electrode-contact resistance compared with a simple two-electrode design.

Before touching a person, validate it using:

- precision resistors;
- resistor-capacitor networks;
- saline phantoms;
- known impedance standards;
- open-circuit and short-circuit tests;
- repeated measurements at different temperatures.

Your first engineering success criterion should be:

"The device accurately recovers the impedance of known test loads across the selected frequency range."

Not:

"The software generates a liver report."

4. Prepare the raw-data format

Never store only "normal," "high" or "low." Save the real measurement.

```

measurement_id
subject_id
  
```



```

device_serial
hardware_revision
firmware_version
timestamp

frequency_hz[]
voltage_real[]
voltage_imag[]
current_real[]
current_imag[]

impedance_real_ohm[]
impedance_imag_ohm[]
magnitude_ohm[]
phase_deg[]

electrode_configuration
contact_quality
skin_temperature
ambient_temperature
measurement_duration

age
sex
height_cm
weight_kg

reference_value
reference_method
reference_timestamp
operator_notes

```

This raw dataset is what you will eventually use to build and audit an AI model.

5. Create a measurement-quality layer

The device must detect bad measurements before making any estimate.

Reject or flag data when there is:

- loose electrode contact;
- excessive movement;
- electrical saturation;
- unstable readings;
- implausible phase or impedance;
- mains interference;
- missing frequencies;
- excessive difference between repeated scans.

A model trained on bad sensor data will learn noise very efficiently—AI is annoyingly talented that way.

6. Define the clinical reference

Every health metric needs an independent source of truth.

For example:

```

Device impedance spectrum
↓
Prediction: total body water
↓
Compare with reference measurement
↓
Calculate error, bias, repeatability and limits of agreement

```



Never use the 21G+ QRMA report as your training label. That would only teach your model to reproduce the existing software's outputs, not actual human physiology.

For every target, specify:

- reference instrument or laboratory method;
- time allowed between sensor and reference measurements;
- participant preparation;
- fasting or hydration requirements;
- posture;
- electrode location;
- room temperature;
- repeated measurements;
- exclusion criteria.

7. Prepare the study protocol before collecting human data

Do not simply collect measurements from friends and start training.

Your protocol should define:

Research question

Participant population

Sample size calculation

Inclusion/exclusion criteria

Reference measurement

Measurement procedure

Risks and safety controls

Consent process

Privacy and data retention

Statistical analysis

Stopping criteria

Adverse-event procedure

Research involving human subjects may require review by an Institutional Review Board or ethics committee. IRBs exist to review the study’s ethics, regulatory compliance and protection of participants. HHS.gov +1

For a significant-risk investigational device study in the United States, FDA and IRB approval may be required before beginning; nonsignificant-risk studies generally still require IRB review. U.S. Food and D... +1

8. Start risk management immediately

Create a risk register before human testing.

Hazard	Possible harm	Initial control
Excessive applied current	Shock or tissue stimulation	Hardware current limiting and isolation
Leakage from USB charger	Electrical injury	Medical-grade isolation or battery operation
Incorrect health result	Delayed medical care	Research-only labeling and no diagnosis
Electrode allergy	Skin irritation	Biocompatible electrodes and screening
Corrupted data	Incorrect prediction	Checksums, validation and audit logs
Model bias	Poor performance in some groups	Representative data and subgroup testing
Personal-data leakage	Privacy harm	Encryption and access control

ISO 14971 defines a lifecycle process for identifying, evaluating, controlling and monitoring medical-device risks. ISO

If this becomes medical electrical equipment, IEC 60601-1 addresses basic safety and essential performance, while IEC 60601-1-2 covers electromagnetic compatibility. Home-use equipment may also need the home-healthcare requirements of IEC 60601-1-11. IEC Webstore +2

9. Do not use deep learning first

Start with interpretable signal processing:

```

Raw spectrum
↓
Noise and contact filtering
↓
Calibration correction
↓
Equivalent-circuit fitting
↓
Physical features
↓
Linear regression / Elastic Net
↓
Validation against reference

```

Useful initial features include:

- resistance at selected frequencies;
- reactance peak;
- phase angle;
- characteristic frequency;
- low- and high-frequency resistance;
- fitted Cole-model parameters;
- repeatability and contact-quality scores.

Only move to XGBoost, neural networks or transformers when you have:

- enough independent participants;
- stable hardware;
- a reliable reference measurement;
- a locked test dataset;
- evidence that simpler models are insufficient.

FDA's Good Machine Learning Practice principles emphasize representative datasets, separation of training and test sets, human-AI interaction, clinically relevant testing and lifecycle monitoring. [U.S. Food and D...](#) +1

10. Set up your development records from day one

Prepare these folders before coding:

```

device-project/
├─ 01-intended-use/
├─ 02-user-needs/
├─ 03-system-requirements/
├─ 04-risk-management/
├─ 05-hardware-design/
├─ 06-firmware/
├─ 07-software/
├─ 08-verification/
├─ 09-clinical-protocol/
├─ 10-datasets/
├─ 11-model-development/
├─ 12-validation/
├─ 13-cybersecurity/
└─ 14-regulatory/

```

Medical-device quality systems require traceability from user needs through design, verification, validation and risk controls. The FDA's Quality Management System Regulation became effective on February 2, 2026 and incorporates ISO 13485:2016 by reference for covered manufacturers. [U.S. Food and D...](#) +1

IEC 62304 provides lifecycle processes for medical-device software. [IEC Webstore](#) +1

Your first prototype

A sensible V0.1 would be:

A research-only four-electrode multi-frequency bioimpedance analyzer that records calibrated complex impedance spectra and estimates hydration trends.

It should initially produce only:

Raw impedance spectrum
Measurement quality score
Resistance/reactance/phase
Equivalent-circuit parameters
Repeated-measurement comparison



Do not initially produce:

Tuberculosis
Cancer
Liver disease
Vitamin deficiency
Blood glucose
50-organ health report
Treatment recommendation



First four deliverables

- 1. Intended-use and single-metric specification
- 2. Bench prototype tested against known R/C loads
- 3. Raw-data acquisition software and database schema
- 4. Human-study protocol with a recognized reference measurement

Once those four are solid, the AI model becomes the relatively easy part.

Sources

Wed, Jun 24 at 4:44 AM

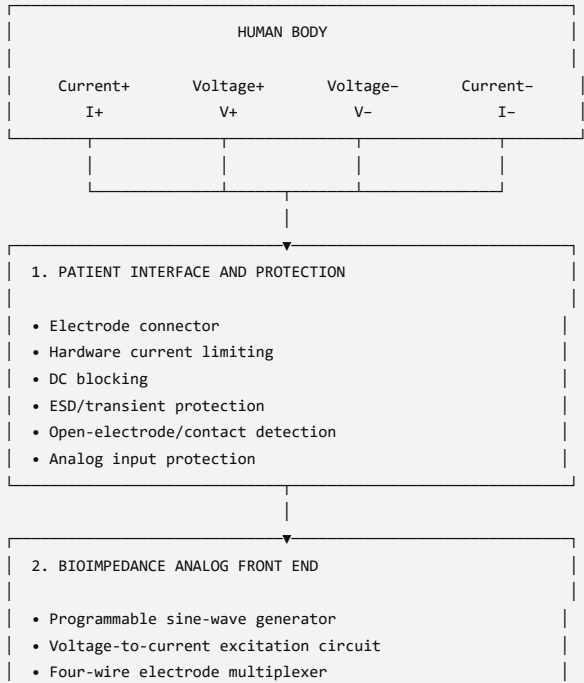
can you give me the architecture for the device design?

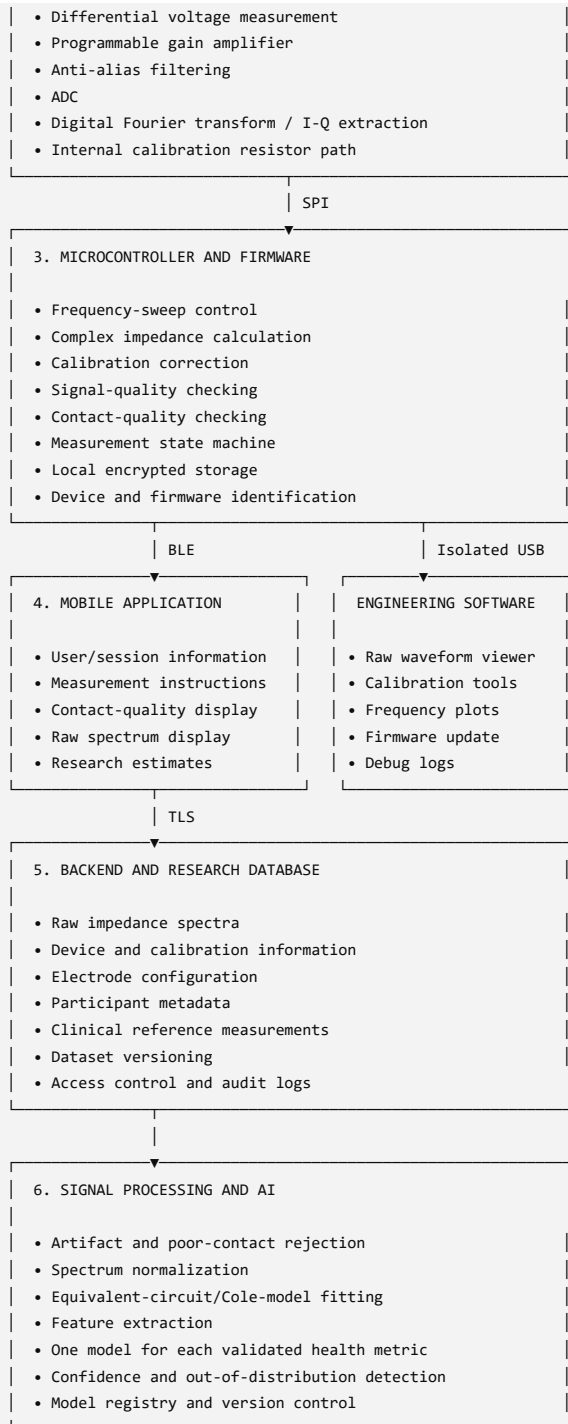
Thought for 50s >

Recommended device architecture

For your first version, build a **multi-frequency bioimpedance spectroscopy device** using four electrodes. Do not design it as a “quantum magnetic resonance” or NMR device unless you are actually building an NMR spectrometer.

This is a **system-level architecture**, not a patient-ready electrical schematic.





1. Electrode architecture

Use a **tetrapolar four-electrode arrangement**:

- **I+** and **I-** inject a small alternating measurement signal.
- **V+** and **V-** measure the resulting voltage.
- The firmware calculates complex impedance from voltage and current.

This is preferable to measuring with only two electrodes because the current-injection and voltage-sensing paths are separated, reducing the effect of electrode contact impedance.

For your first prototype, use standard disposable gel electrodes rather than custom metal hand plates. They give you better control over:

- position;
- contact area;

- skin preparation;
- contact stability;
- repeated measurements.

Choose one measurement path initially, such as:

Right wrist → left wrist



or:

Right hand → right foot



Do not switch between several body paths until the first one is repeatable.

2. Analog front-end selection

Recommended: AD5940 or AD5941

The AD5940/AD5941 is the better starting point for a research frequency-sweep device. It is designed for skin and body impedance applications, supports an absolute four-wire body-impedance measurement, has a 16-bit 800 kSPS ADC, an on-chip DFT accelerator, and a high-bandwidth excitation path supporting measurements up to 200 kHz. [Analog Devices +2](#)

Use it approximately like this:

```

AD5940 waveform generator
↓
External protection/current-limit network
↓
I+ and I- electrodes
↓
V+ and V- sense electrodes
↓
AD5940 differential measurement path
↓
ADC + DFT
↓
Real and imaginary impedance results
  
```



A reasonable research sweep could contain logarithmically spaced frequencies such as:

```

5 kHz
10 kHz
20 kHz
30 kHz
50 kHz
75 kHz
100 kHz
150 kHz
200 kHz
  
```



The excitation amplitude and patient-leakage protection must be established through formal electrical-safety analysis and testing, not copied casually from a hobby circuit.

Alternative: AFE4300

The TI AFE4300 is more appropriate when the primary application is a weighing scale with body-composition measurement. It combines weight-scale circuitry with support for up to three tetrapolar complex-impedance measurements. [Texas Instruments +1](#)

My recommendation:

```

Research spectrum analyzer → AD5940/AD5941
Smart body-composition scale → AFE4300
  
```



3. Microcontroller architecture

Use an MCU such as:

- an STM32U5 or STM32L4 for straightforward embedded development;
- an nRF52840 if Bluetooth Low Energy is important;
- an ESP32 only for an early bench prototype, not as the final patient-connected architecture without careful isolation and noise evaluation.

The MCU firmware should follow a controlled state machine:

```

POWER_ON
↓
SELF_TEST
↓
CALIBRATE
↓
WAIT_FOR_ELECTRODES
↓
CHECK_CONTACT
↓
RUN_FREQUENCY_SWEEP
↓
VALIDATE_SIGNAL
↓
CALCULATE_IMPEDANCE
↓
STORE_RAW_RESULTS
↓
TRANSFER_RESULTS
↓
COMPLETE

```



The MCU should calculate, for every frequency:

```

frequency
voltage_real
voltage_imaginary
current_real
current_imaginary
impedance_real
impedance_imaginary
impedance_magnitude
phase_angle
signal_quality

```



The basic complex calculation is:

$$Z(f) = V(f) / I(f)$$


Do not initially calculate “liver health,” “vitamin condition,” or “blood sugar.” First verify that the device accurately measures known electrical impedances.

4. Calibration architecture

Include a calibration path inside the hardware:

```

Analog multiplexer
├─ Human electrodes
├─ Precision calibration resistor
├─ Open-circuit test
└─ Internal diagnostic path

```



Create an external calibration fixture containing:

- precision resistors;
- resistor-capacitor networks;
- several expected body-impedance ranges;
- open and short conditions;
- optional saline phantoms.

Before any human measurement, the device should pass:

DC offset test
Gain test
Phase-error test
Frequency accuracy test
Known-resistance test
Known R-C impedance test
Open-electrode test
Noise-floor test



The calibration coefficients should be tied to:

device serial number
hardware revision
AFE lot or version
firmware version
calibration date
temperature
calibration fixture ID



5. Power and patient-safety architecture

During measurement, use battery power.

```

Li-ion battery
  ↓
Battery protection
  ↓
Low-noise analog regulator → Analog front end
  ↓
Digital regulator → MCU and Bluetooth
  
```



Essential protection mechanisms should include:

- independent hardware current limiting;
- DC blocking;
- measurement timeout;
- watchdog-controlled shutdown;
- overvoltage protection;
- electrode-open detection;
- fault-state disconnection;
- separation of analog and noisy radio power domains.

Do not permit a direct, non-isolated USB charger or desktop-PC ground connection while the user is attached to the electrodes. Use isolated USB for development, or physically prevent simultaneous USB connection and human measurement.

IEC 60601-1 covers basic safety and essential performance for medical electrical equipment, while IEC 60601-1-2 covers electromagnetic compatibility and immunity. A home-use product may also fall under the home-healthcare requirements of IEC 60601-1-11. [IEC Webstore +2](#)

6. PCB architecture

For the first custom PCB, separate it into functional zones:

```

| Patient connectors |
| Protection + current-limiting area |
| Quiet analog area: AFE, references, filters |
| Isolation boundary |
| Digital area: MCU, memory, BLE |
  
```



Battery, charger and power management

Important layout rules:

- Keep the AFE, reference resistors and electrode inputs close together.
- Keep Bluetooth, switching regulators and clocks away from high-impedance analog inputs.
- Use separate analog and digital power filtering.
- Add guarded high-impedance traces where needed.
- Provide test points for excitation, measured voltage, references and power rails.
- Add a shielded enclosure option for laboratory testing.

7. Mobile application architecture

The first application should show measured data, not generate dozens of health diagnoses.

```

Device connection
↓
Measurement instructions
↓
Electrode/contact validation
↓
Live measurement progress
↓
Impedance magnitude versus frequency
↓
Phase versus frequency
↓
Quality score
↓
Upload to research database
  
```



Initial user-visible outputs:

```

Measurement succeeded/failed
Contact quality
Resistance and reactance
Phase angle
Spectrum graph
Repeatability compared with previous scan
Research-only metric estimate
Confidence interval
  
```



8. Backend architecture

Use separate storage for raw measurements and derived results:

```

Raw Measurement Store
├─ complete frequency spectra
├─ calibration details
├─ device metadata
└─ signal-quality information

Clinical Reference Store
├─ reference method
├─ measured value
├─ laboratory/instrument ID
└─ measurement timestamp

Derived Feature Store
├─ Cole-model parameters
├─ phase features
├─ resistance features
└─ quality features

Model Store
├─ model version
├─ training dataset version
└─ target metric
  
```



```
└─ validation results
└─ deployment status
```

A useful measurement record could be:

```
{
  "device_serial": "BIOZ-0001",
  "hardware_revision": "A",
  "firmware_version": "0.1.0",
  "electrode_configuration": "right_wrist_left_wrist",
  "frequencies_hz": [5000, 10000, 20000, 50000, 100000],
  "resistance_ohm": [],
  "reactance_ohm": [],
  "phase_deg": [],
  "contact_quality": {},
  "skin_temperature_c": null,
  "reference_test": {
    "type": "validated_reference_method",
    "value": null,
    "measured_at": null
  }
}
```

JSON



9. AI architecture

Do not use one giant model to generate 50 health reports.

Use a staged pipeline:

```
Raw complex impedance spectrum
↓
Signal-quality classifier
↓
Artifact/contact rejection
↓
Calibration and normalization
↓
Physical feature extraction
↓
Cole-model/equivalent-circuit fitting
↓
Target-specific prediction model
↓
Uncertainty and validity check
↓
Research estimate or "unable to estimate"
```



The AI should contain at least four separate components:

Measurement-quality model

Determines whether the scan is usable.

Feature extractor

Produces:

- low-frequency resistance;
- high-frequency resistance;
- reactance peak;
- characteristic frequency;
- phase-angle features;
- fitted equivalent-circuit parameters;
- repeatability features.

Target-specific model

One model for each scientifically validated endpoint:

Model A → hydration estimate
 Model B → extracellular-water estimate
 Model C → body-composition estimate



Uncertainty detector

Rejects measurements outside the validated population or signal range.

The system must be able to return:

Insufficient signal quality
 Unsupported participant profile
 Electrode contact unstable
 Measurement outside validated range
 No result produced



That is far safer than forcing a confident-looking number from bad data.

10. Best first prototype

Do not create a custom PCB immediately.

Start with:

EVAL-AD5940BIOZ evaluation board
 ADuCM3029 evaluation MCU board
 Four disposable electrodes
 Battery-powered laptop interface or isolated USB
 Precision R/C calibration board
 Basic raw-data acquisition program



Analog Devices specifically provides the EVAL-AD5940BIOZ kit for bioimpedance/BIA and publishes example software for body-impedance and electrodermal-activity applications. [Analog Devices](#) +1

Your development sequence should be:

Phase 1: Measure known resistors
 Phase 2: Measure known R-C networks
 Phase 3: Measure saline phantoms
 Phase 4: Verify repeatability on humans
 Phase 5: Collect paired clinical-reference data
 Phase 6: Build one prediction model
 Phase 7: Independent validation
 Phase 8: Custom PCB



Recommended V0.1 scope

Device:
 Four-electrode multi-frequency bioimpedance analyzer

Target:
 Raw calibrated complex impedance spectrum

First derived metric:
 Hydration or body-water trend

Output:
 Research-only estimate with quality and confidence

Not included:
 Disease diagnosis
 Tuberculosis detection
 Cancer detection
 Vitamin diagnosis
 Treatment recommendations
 Universal organ reports



If the device is eventually marketed for medical diagnosis in the United States, its development process will also need to fit the FDA Quality Management System Regulation, which has been effective since February 2, 2026 and incorporates ISO 13485:2016 by reference. [fda.gov +1](#)

    ...   Sources

okay, if I build the v0.1 scope, then what's next?

Thought for 35s >

After V0.1, the next goal is **not adding more health reports**. The next goal is proving that V0.1 measures impedance accurately, safely, and repeatedly.

Development roadmap



V0.2 — Calibrated engineering instrument

V0.2 should prove the electronics, without attempting to measure health.

Add these components

- Internal precision calibration resistor
- External resistor/RC calibration fixture
- Open-circuit and short-circuit diagnostics
- Temperature monitoring
- Electrode-contact detection
- Device self-test
- Raw spectrum export
- Calibration coefficient storage
- Hardware and firmware version tracking

Test the device against:

- precision resistors;
- known resistor-capacitor networks;
- saline phantoms;
- open circuits;
- short circuits;
- multiple temperatures;
- repeated power cycles.

Suggested internal targets

These are engineering targets, not regulatory limits:

Resistance magnitude error:	≤ 1-2%
Phase error:	≤ 1-2°
Repeated phantom variation:	≤ 1%

Frequency accuracy: $\leq 0.1\%$
 No unexplained drift over repeated scans

Exit gate

Do not proceed until this statement is true:

The device can accurately recover known complex impedances across its complete frequency range.

V0.3 — Human repeatability prototype

Now test whether measurements are stable on people. Do not test disease detection yet.

Standardize:

Electrode type
 Electrode position
 Skin preparation
 Participant posture
 Rest period
 Measurement duration
 Room temperature
 Food and drink restrictions
 Exercise restrictions
 Time of day



For every participant, take repeated scans:

Scan 1
 Wait briefly
 Scan 2
 Remove and replace electrodes
 Scan 3
 Repeat on another day



Evaluate:

- within-session repeatability;
- between-session repeatability;
- effect of replacing electrodes;
- effect of posture;
- effect of hydration;
- effect of movement;
- contact-quality failures;
- differences between device units.

The output should still be:

Resistance
 Reactance
 Phase
 Spectrum
 Cole-model parameters
 Signal-quality score



Not organ diagnoses.

Human research normally needs an ethics/IRB review appropriate to the study. In the United States, significant-risk device studies require FDA and IRB approval; nonsignificant-risk studies generally require IRB approval under abbreviated IDE requirements. The risk determination should be documented rather than assumed. [fda.gov](https://www.fda.gov) +1

V0.4 — Build a reference-linked dataset

Choose **one health metric**.

A good initial target could be:

- total body water;
- extracellular-water ratio;
- hydration trend;
- body composition;
- localized edema trend.

For each participant, acquire the impedance measurement and an accepted reference measurement close together.

Your device measurement
+
Reference measurement
+
Participant metadata
+
Measurement conditions
=
Training record



Example record:

```
{
  "subject_id": "anonymous-001",
  "device_serial": "BIOZ-0003",
  "firmware_version": "0.4.1",
  "electrode_path": "right_hand_right_foot",
  "frequency_hz": [5000, 10000, 20000, 50000, 100000],
  "resistance_ohm": [],
  "reactance_ohm": [],
  "phase_deg": [],
  "contact_quality": 0.94,
  "posture": "supine",
  "rest_minutes": 10,
  "height_cm": 175,
  "weight_kg": 72,
  "reference_type": "validated_reference_method",
}
```

JSON



Important rule:

Do not use QRMA-generated reports as reference labels.

Otherwise, your model will only imitate the old QRMA software.

V0.5 — Build the first prediction model

Begin with an interpretable model:

Raw impedance spectrum
↓
Quality rejection
↓
Calibration correction
↓
Feature extraction
↓
Cole-model fitting
↓
Linear regression / Elastic Net
↓
Target estimate and uncertainty



Only later compare it against:

- random forest;
- gradient-boosted trees;

- neural networks.

Data splitting

Split by participant:

Training participants
Validation participants
Test participants



Never put scans from the same person in both training and testing. That can make the reported accuracy look much better than real-world performance.

Keep the final test set untouched until the model, features, and thresholds are frozen.

The current IMDRF/FDA Good Machine Learning Practice principles emphasize representative datasets, separation of training and test data, clinically relevant evaluation, human-AI interaction, transparency, and monitoring across the product lifecycle. [fda.gov](#) +1

Model output

Use:

Estimated total body water: 38.2 L
Expected error range: ± 2.4 L
Measurement quality: Good
Validated population: Adults 18-65



Allow the model to refuse:

No estimate produced:
electrode contact was unstable.



V0.6 — Independent validation

After model development, validate it using:

- new participants;
- new device units;
- different operators;
- preferably another testing location;
- the same locked firmware and model.

Measure:

Mean absolute error
Bias
Limits of agreement
Test-retest variation
Failure rate
Calibration error
Sensitivity to electrode placement
Performance by relevant subgroups



Do not modify the model using the independent-validation dataset. If you modify it, that dataset becomes development data and you need another independent test.

V0.7–V0.9 — Convert research hardware into a product

Once the measurement and model work, redesign the hardware.

Product hardware

Custom PCB
Battery operation
Independent current limiting
Electrical isolation



Protected electrode interface
 Production calibration connector
 Secure boot
 Signed firmware
 Watchdog
 Enclosure
 Production test points

Product software

Firmware state machine
 Mobile application
 Secure device pairing
 Encrypted data transport
 User management
 Audit logging
 Model versioning
 Remote update controls
 Data deletion and export



Design documentation

User needs
 Intended use
 System requirements
 Electrical requirements
 Software requirements
 Risk analysis
 Architecture
 Verification protocols
 Validation protocols
 Traceability matrix
 Manufacturing tests
 Cybersecurity plan
 Post-market monitoring plan



ISO 14971 applies risk management throughout the device lifecycle, including electrical, software, cybersecurity and usability risks. ISO

IEC 62304 defines lifecycle processes for medical-device software and applies to embedded software as well as software that is itself a medical device. IEC Webstore

For electrodes and other patient-contacting materials, biocompatibility should be assessed based on the material, manufacturing process, contact location, and contact duration. fda.gov +1

V1.0 — Regulatory and manufacturing preparation

Before spending heavily on a large clinical study, define your intended claim:

Lower-risk research/wellness claim

Measures and displays multi-frequency impedance and hydration trends.

Medical claim

Detects abnormal fluid status or supports diagnosis of a medical condition.

That wording changes the likely regulatory pathway and evidence requirements.

For a U.S. medical-device product, an FDA Pre-Submission can be used to request feedback on intended use, device classification, testing plans, clinical protocol and proposed submission strategy before completing expensive studies. fda.gov

As of February 2, 2026, the FDA Quality Management System Regulation incorporates ISO 13485:2016 by reference for covered device manufacturers. Product development should therefore establish design, supplier, manufacturing, corrective-action and document-control processes before commercial release—not afterward. fda.gov

Your immediate next work package

After V0.1 is assembled, complete these five tasks in order:

1. Build the R/C calibration fixture
2. Write automated calibration and repeatability tests
3. Add contact-quality and fault detection
4. Freeze the raw-data format
5. Produce a V0.2 verification report



Your V0.2 report should answer:

What frequencies were tested?
 What reference loads were used?
 What was the magnitude error?
 What was the phase error?
 How repeatable were the results?
 How did temperature affect the result?
 How were open and loose electrodes detected?
 Did multiple device units produce equivalent measurements?



Only after those questions have convincing answers should you begin collecting the health-reference dataset.

can you give me the block diagram for the v0.1 device?
 program code as well

I prepared a **V0.1 bench-only starter package** containing the firmware wrapper, PC capture/plotting program, sample data, and a text block diagram:

[Download the BioZ V0.1 starter package](#)

The package includes:

```

bioz_v01_starter/
├── firmware/
│   ├── AD5940Main_v01.c
│   └── v01_config.h
├── host/
│   ├── capture_and_plot.py
│   └── requirements.txt
├── sample/
│   └── rc_sweep.csv
├── docs/
│   └── block_diagram.md
└── README.md
  
```



The architecture is:

```

Known R/C test load
    ↓
EVAL-AD5940BIOZ
    ↓ SPI
EVAL-ADICUP3029 MCU
    ↓ UART/USB
PC capture and plotting program
  
```



The firmware performs a logarithmic **5 kHz–100 kHz sweep**, calculates magnitude, phase, real impedance, and imaginary impedance, then outputs CSV records. It is designed to plug into Analog Devices' official `AD5940_BIA` example and depends on its `BodyImpedance.c`, `BodyImpedance.h`, `ad5940lib`, and MCU port files. Analog Devices provides the official BIA example and evaluation platform for this purpose.

One important correction: this V0.1 must be tested only with the supplied impedance test board or known resistor/capacitor loads. Analog Devices explicitly warns not to connect the AD5940 evaluation board directly to a human body. A human-connected prototype needs a separately engineered and verified patient-protection design.

The host program can read the sample immediately:

</> Bash

For a serial device:

```
python host/capture_and_plot.py \  
--port COM5 \  
  
</> Bash
```

</>

Thought for 43s >

Stopped creating image

    ...  Sources

where can I find the AI model or database for these health measurement record

Thought for 43s >

For the **V0.1 AD5940 bioimpedance device**, the best public starting database is **CDC NHANES Bioelectrical Impedance Analysis data**.

There is no credible plug-and-play AI model that accepts your impedance spectrum and produces the 50 health reports claimed by QRMA products. You will need a **target-specific model**, such as body water, fat-free mass, or respiratory monitoring.

1. Best immediate dataset: NHANES BIA

NHANES released downloadable BIA datasets for survey cycles including 1999–2000, 2001–2002, 2003–2004, and 2005–2006. The files are public SAS XPT datasets.  CDC +2

The 2001–2002 data contain resistance and reactance measurements across many frequencies from **5 kHz to 1 MHz**, including values matching your V0.1 sweep:

Your measurement	NHANES field example
Resistance at 5 kHz	BIXS005K
Reactance at 5 kHz	BIXC005K
Resistance at 10 kHz	BIXS010K
Reactance at 10 kHz	BIXC010K
Resistance at 20 kHz	BIXS020K
Reactance at 20 kHz	BIXC020K

Your measurement	NHANES field example
Resistance at 50 kHz	BIXS050K
Reactance at 50 kHz	BIXC050K
Resistance at 100 kHz	BIXS100K
Reactance at 100 kHz	BIXC100K
Resistance at 200 kHz	BIXS200K
Reactance at 200 kHz	BIXC200K

NHANES also includes calculated Cole-model and body-composition fields such as:

BIDALPHA	Cole-model alpha	
BIDFC	Characteristic frequency	
BIDRECF	Extracellular-fluid resistance	
BIDRICF	Intracellular-fluid resistance	
BIDTBW	Estimated total body water	
BIDECF	Estimated extracellular fluid	
BIDICF	Estimated intracellular fluid	
BIDFAT	Estimated fat mass	
BIDFFM	Estimated fat-free mass	
BIDPFAT	Estimated body-fat percentage	

These variables are documented in the official NHANES variable list.

Use `SEQN`, the participant identifier, to join BIA data with:

DEMO	demographics	
BMX	height, weight, waist and other body measurements	
DXX	DXA body-composition measurements where available	
LAB	laboratory measurements	
BPX	blood pressure	

NHANES 1999–2000 provides both BIA and whole-body DXA data, making it especially useful for studying calibration of impedance-derived body composition against a stronger reference method.

Important limitation

Fields such as `BIDTBW`, `BIDFAT`, and `BIDFFM` are themselves estimates calculated from impedance. Training an AI model only against those fields teaches it to imitate an existing equation.

For better validation:

Impedance spectrum → predict DXA fat mass	
Impedance spectrum → predict DXA lean mass	

is stronger than:

Impedance spectrum → predict BIA-derived fat mass	
---	--

2. UK Biobank: larger but controlled access

UK Biobank collected body impedance, calculated body-composition measurements, anthropometrics, biomarkers, medical records, and other health data. Its protocol states that both measured raw bioimpedance values and calculated values were captured. +1

It is potentially valuable for large-scale model development, but it is not an unrestricted public download. Eligible researchers must submit a health-related research application and work through UK Biobank’s controlled Research Analysis Platform. +2

A UK Biobank analysis comparing BIA with DXA included 34,437 participants. It found strong population-level correlation but meaningful individual-level disagreement, which is precisely why your model needs individual validation and uncertainty reporting. 

3. Zenodo bioimpedance spectroscopy collection

The Zenodo dataset “**Single dispersion EIS measurements: a compilation**” contains impedance spectroscopy measurements from several instruments, with plant, phantom, and human-tissue measurements collected across multiple laboratories. 

This is useful for:

- testing spectral import code;
- Cole-model fitting;
- comparing instruments;
- validating preprocessing;
- classifying phantom versus tissue spectra.

It is less useful as a direct health-prediction dataset because it does not provide a large standardized set of clinical labels.

4. PhysioNet for respiratory impedance applications

PhysioNet has datasets containing raw electrical impedance tomography data together with respiratory pressure, flow, ECG, and heart-rate measurements. One current dataset includes 20 subjects and paired respiratory measurements. 

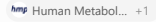
Use this only if your goal becomes:

- respiratory-rate estimation;
- lung aeration monitoring;
- breathing-pattern classification.

EIT uses many electrodes and image reconstruction, so its data are not directly interchangeable with your four-electrode whole-body BIS hardware.

5. NMR databases are for a different device

If you eventually build a **real NMR spectrometer**, useful databases include:

- **HMDB**, which supports searches of experimental human-metabolite NMR spectra;  +1
- **MetaboLights**, which stores public metabolomics studies, raw experimental files, reference spectra, biological roles, and concentrations. 

These cannot train the current AD5940 bioimpedance device. NMR chemical-shift spectra and electrical impedance spectra are physically different measurements.

AI model recommendation

For V0.1, do not search for a deep-learning model first. Build this baseline:

```
Resistance and reactance at each frequency
↓
Measurement-quality filtering
↓
Cole-model fitting
↓
Feature extraction
↓
Elastic Net regression
↓
DXA fat mass or lean mass
```



Input features could include:

```
R_5k, X_5k
R_10k, X_10k
R_20k, X_20k
```



R_50k, X_50k
R_100k, X_100k
R_200k, X_200k

height
weight
age
sex

Cole_R0
Cole_Rinf
Cole_alpha
Cole_characteristic_frequency
phase_angle_50k

Then compare:

Elastic Net
Random Forest
XGBoost
Small neural network



Split the data by participant, never by scan.

Recommended development sequence

1. Download NHANES BIX, BMX, DEMO and DXA files
2. Merge them using SEQN
3. Select frequencies supported by your AD5940 device
4. Train a body-composition baseline using NHANES
5. Measure known R/C loads with your V0.1 device
6. Collect your own paired impedance + reference dataset
7. Retrain or calibrate the model for your hardware
8. Validate it on completely new participants and device units



The most practical first target is **DXA-referenced fat-free mass or fat mass**. Hydration is also possible, but it requires an independent accepted reference method; training against BIDTBW alone would only reproduce NHANES’s existing BIA calculation.

Sources

Sun, Jun 28 at 9:23 AM

I am looking for the data for MetaTrone which is a kind of a QRMA

Thought for 32s >

You probably mean **Metatron NLS**, often used with software called **Metapathia**, **Metapathia GR**, or **Metapathia Hospital**. It is related to the QRMA/bioresonance market, but it is a different hardware and software family from the 21G+ palm-sensor analyzer.

Where the Metatron data is stored

Metatron appears to use three different kinds of data:

Data type	What it contains	Publicly available?
Etalon/reference database	Claimed reference patterns for organs, diseases, microorganisms, allergens, medicines and other categories	No public raw download found
Patient/session database	Names, examinations, reports, screenshots and previous scan results	Stored locally by the licensed software

Data type	What it contains	Publicly available?
Raw device signal	Data received from the headset/device before comparison with references	No public protocol or documented raw format found

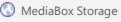
The official Metatron site says its software contains **more than 12,000 “reference etalons”**, including virtual organ models, biochemical homeostasis, nosology, bacteria, viruses, fungi, allergens, medicines and nutritional products. However, the site provides descriptions of those categories—not the underlying numerical spectra, file schema, model weights or training records. 

The Metapathia Hospital product page similarly describes additional etalon databases bundled with the software, but does not offer them as CSV, SQL, JSON or another open research format. 

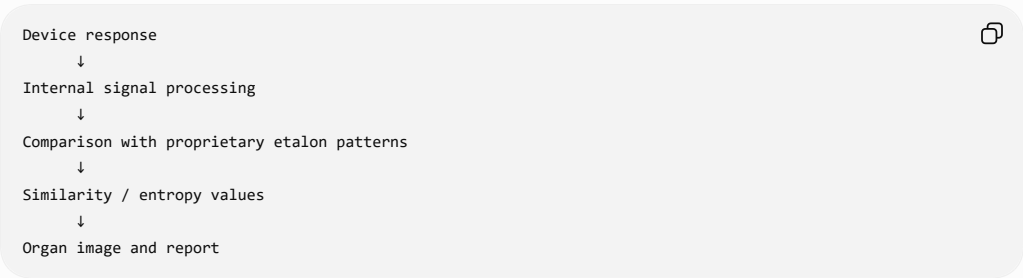
Is there an AI model?

I found no public Metatron AI model containing:

- model weights;
- training code;
- training patients;
- raw frequency vectors;
- clinically verified labels;
- an API specification;
- an etalon data dictionary.

Public product material describes a **similarity-comparison system**: the measured result is compared with stored etalons and displayed in decreasing order of “spectral similarity.” One public Metatron brochure shows an example list with reference items and numerical similarity values, but not the underlying raw reference curves. 

So it appears closer to:



rather than a modern published machine-learning model with downloadable weights.

How to obtain the legitimate database

The proper source is the **Institute of Practical Psychophysics**, the developer of the original HSS Metatron product. Their official site lists product support and contact information and states that the etalon base is continually expanded. 

Ask them or an authorized distributor specifically for:

1. Licensed Metapathia software and hardware compatibility
 2. Etalon database licence
 3. Data dictionary for the 12,000 reference etalons
 4. Raw examination export format
 5. Raw sensor or waveform export
 6. Device-to-PC communication protocol
 7. SDK or API documentation
 8. Anonymized sample examinations
 9. Calibration method
 10. Clinical reference used to create each etalon
- 

A useful request would be:

We are developing a research bioimpedance/bioresonance system and need licensed technical access to the Metatron reference-etalon database. Please provide information about database licensing, etalon record structure, raw scan export, sensor-data format, API/SDK availability, calibration documentation and the clinical reference data used to create the etalons.

Avoid random "Metatron database downloads" and cracked Metapathia packages. They may contain malware, modified reference data or incompatible dongle protection, and they would not provide trustworthy provenance.

When you already own a licensed Metatron system

The most useful next step is to inspect the installed software without changing it. In its installation folder, run this PowerShell command:

```
Get-ChildItem -Recurse -File |  
Group-Object Extension |
```

PowerShell



Then list likely data files:

```
Get-ChildItem -Recurse -File |  
Where-Object {  
    $_.Extension -in ".db", ".sqlite", ".mdb", ".accdb",  
        ".dat", ".bin", ".xml", ".json",  
        ".csv", ".ini"  
} |
```

PowerShell



This can reveal whether the licensed installation uses SQLite, Access, binary resource files or encrypted proprietary packages. It will not prove which files contain etalons, but it gives us a safe starting inventory.

Important limitation

Even if you obtain the 12,000 etalons, they should not automatically become the training truth for your new device. The vendor database could help you reproduce Metatron-style outputs, but it would not by itself establish that those outputs correspond to laboratory measurements or medically confirmed diagnoses.

For a scientifically defensible model, you need records like:

```
raw sensor spectrum  
+ device calibration  
+ electrode/headset configuration  
+ subject conditions  
+ accepted clinical test result  
= validated training record
```

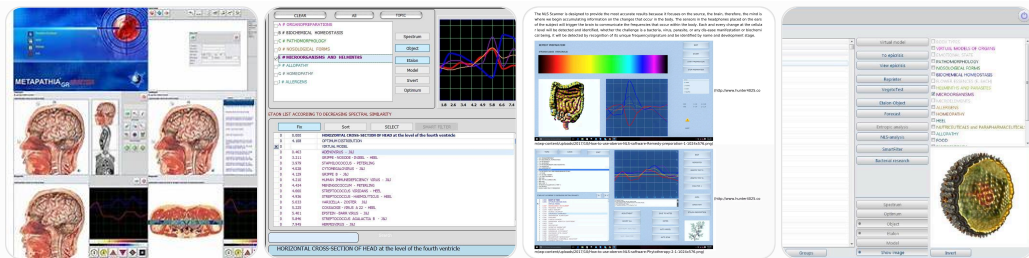


The best practical route is to obtain a licensed Metatron installation, export several anonymized examinations, and compare its raw files before deciding whether the data can be reused.

Sources

give me the screen for the metatron NLS and UI

Thought for 19s >



These are representative screens from the **Metatron NLS / Metapathia** software family. The exact layout varies between **Metapathia GR Hunter**, **Metapathia GR Clinical**, and **Metapathia Hospital**. [Metatron +2](#)

The usual UI contains:

- **Patient management screen** — create or select a patient and open previous examinations.
- **Anatomical model viewer** — displays 2D, pseudo-3D, or 3D images of organs and tissue sections.
- **Spectrum graph** — shows several colored curves representing the selected object and reference “etalon.”
- **Etalon database browser** — categories such as organ models, biochemical homeostasis, pathomorphology, microorganisms, allergens, homeopathy, food, and medicines.
- **Similarity-results table** — lists reference items ordered by the software’s reported spectral similarity.
- **Analysis controls** — buttons such as **Spectrum**, **Object**, **Etalon**, **Entropy analysis**, **NLS analysis**, **Smart Filter**, and **VegetoTest**.
- **Report module** — generates examination protocols and comparison reports.

A simplified layout looks like this:

Patient Examination Settings Report Exit				
Anatomical tree	Anatomical organ image with numbered markers		Controls	
• Head • Brain • Heart • Liver • Lungs			- Spectrum - Object - Etalon - Analysis - Report	
Etalon categories	Colored spectral comparison graph			
Reference results ordered by reported similarity				
Value	Name	Category	Similarity	Notes

These screenshots show how the proprietary software presents anatomy, reference libraries, and spectral-comparison results. They do not demonstrate that the measurements or medical interpretations are clinically accurate.

[Sources](#)

I'd like to see the picture of the device of metaron NLS or Metapathia GR Hunter, Metapathia GR Clinical, Metapathia Hospital

Thought for 12s >



These are representative photos of the **Metatron NLS / Metapathia GR Hunter** hardware. The normal kit typically includes:

- A compact metal controller/interface box
- A headset-like pair of magnetic inductors or "trigger sensors"
- USB and sensor cables
- A Windows computer running the Metapathia software
- Sometimes a carrying case, hand sensor, or metal test cup

The device in the second and fourth photos shows the usual operating setup: the headset connects to the controller box, which connects to a computer displaying the Metapathia interface. [MediaBox Storage +1](#)

Differences between the names

Name	Usually refers to
Metatron NLS	The overall hardware-software system
Metatron 4025 / 4025M	Common controller hardware models
Metapathia GR Hunter	Software version commonly sold with the 4025 hardware
Metapathia GR Clinical	A more professionally marketed software package
Metapathia Hospital	A newer software edition based on GR Clinical, generally using similar Metatron hardware

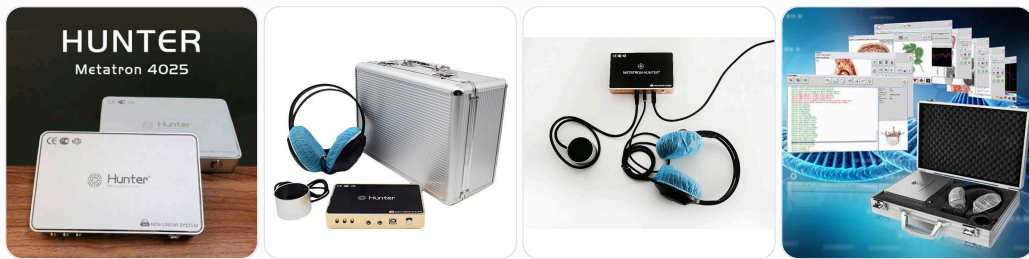
Metapathia GR Clinical and Metapathia Hospital are mainly software editions, so the physical device may look nearly identical. The official product information says Hospital is based on GR Clinical and includes its functions and reference libraries. [Метатрон +1](#)

Be aware that many online photographs are from resellers, and boxes marked "**Meta Hunter**," "**Hunter 4025**," "**25D**," or "**18D**" may be third-party or rebranded products rather than original Institute of Practical Psychophysics hardware. The external appearance alone does not establish authenticity or medical certification.

[Sources](#)

then give me the more detailed for meta hunter and 25D

Thought for 31s >



Meta Hunter 4025 and “25D” explained

The names are often mixed together, but they refer to different parts of the product:

Name	Usually means
Metatron 4025	The hardware–software system/model family
Meta Hunter / Hunter 4025	A commercial or reseller name for the controller and headset package
Metapathia GR Hunter	The Windows software historically associated with the 4025
18D / 25D	Reseller software-edition or generation labels—not physical dimensions
Metapathia GR Clinical	A differently positioned software package
Metapathia Hospital	A later software/library package in the broader product family

The crucial point about “25D”

“25D” does **not** mean that the equipment performs 25-dimensional medical imaging. It is normally a marketing label for a software package said to contain additional anatomical images, databases and analysis functions.

Even the available Hunter walkthrough manual says that the feature marketed as “3D” is actually based on **2D anatomical images with selectable locations and deeper linked images**, rather than genuine three-dimensional medical imaging.  ISHA Bioresonan...

Furthermore, the Institute of Practical Psychophysics’ official Russian site states that versions carrying the **Metapathia GR Hunter** name released after January 1, 2018 are regarded by the institute as non-original products. This is highly relevant because many current marketplace listings use labels such as **18D, 25D**, “2025 edition,” or “latest generation.”  Metatron NLS

Therefore, “25D Metatron” products sold online may be:

- third-party hardware;
- modified or copied software;
- rebranded Hunter-style systems;
- unofficial database expansions;
- generic controller boxes using the Metatron name.

Typical hardware package

A Meta Hunter/25D kit usually contains the following components.

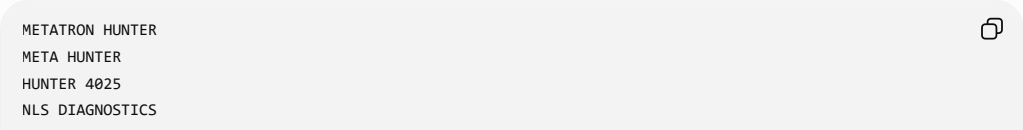
1. Controller box

The small rectangular box normally provides:

- USB connection to the Windows computer;
- headset connection;
- accessory or “cup” connection;
- status indicators;
- internal signal-generation and interface electronics;
- sometimes a separate power input.

The box is marketed as the scanner, but publicly available materials do not provide a reproducible circuit schematic, ADC specification, raw sensor format, calibration procedure or validated measurement protocol.

Different sellers use externally similar boxes with different labels:



25D NLS

18D NLS

Their external similarity does not prove that their internal electronics or software protocols are identical.

2. Headset or “trigger sensors”

The headset resembles ordinary over-ear headphones. Sellers describe the two earpieces as:

- magnetic inductors;
- trigger sensors;
- bioinductors;
- signal receivers/transmitters;
- resonance sensors.

In the operating manual, the software’s Research button becomes active only after the headset is connected and positioned. The manual also distinguishes between a headset connector and a separate cup connector.

 ISHA Bioresonan...

However, public documentation does not adequately disclose:

- whether each earpiece contains a coil, magnet, electrode or active amplifier;
- whether the device measures voltage, magnetic flux, impedance or something else;
- signal amplitude;
- sampling rate;
- frequency response;
- signal-to-noise ratio;
- electrode or coil calibration;
- raw data produced during a scan.

Without these details, it cannot be assumed that the headset is collecting clinically meaningful resonance data.

3. Test cup or “Vegetotest” cup

Many kits include a metal cylindrical cup. In the software workflow, an item such as a supplement or medicine is placed inside the cup and compared with a selected organ or reference pattern.

The Hunter manual describes two kinds of Vegetotest:

- a **virtual test** using entries already stored in the software library;
- a supposedly **physical test** using an item placed in the connected cup.  ISHA Bioresonan...

The manual also describes a “Reprinter” that supposedly transfers stored frequency information into water, alcohol, sugar or another medium. These are product claims; the manual does not supply accepted physical validation showing that such information is transferred or produces a medical effect.  ISHA Bioresonan...

4. USB protection key

A licensed or packaged system may include a USB dongle used for:

- software activation;
- database access;
- hardware identification;
- copy protection;
- version licensing.

A dongle does not necessarily contain the entire etalon database. It may simply authorize access to files installed on the computer.

5. Windows computer

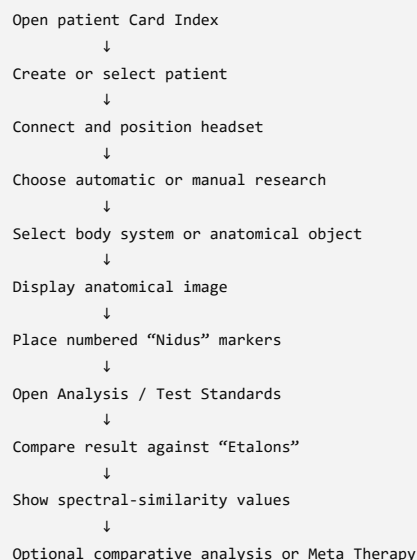
The system generally runs as a Windows desktop application. Marketplace packages frequently ship with a preconfigured laptop because the software may depend on:

- a specific USB driver;
- fixed installation paths;
- an activation dongle;

- older runtime libraries;
- a particular Windows release;
- locale or language settings.

Software workflow

The published Hunter walkthrough describes the following process.  ISHA Bioresonan...



Patient Card Index

The software begins with a local patient database. A record may contain:

- name and demographics;
- examination date;
- selected anatomical objects;
- scan results;
- analysis screens;
- previous and comparative examinations.

Automatic scan

According to the manual, automatic mode begins with the head and proceeds through a collection of anatomical images. Areas selected by the software may be marked with colored indicators.  ISHA Bioresonan...

Manual scan

The walkthrough lists nine investigation groups:

1. Alimentary
2. Respiratory
3. Urogenital
4. Cardiovascular
5. Blood and lymph
6. Endocrine
7. Nervous
8. Sensory systems
9. Locomotor system

The operator selects an image or anatomical item from these categories.  ISHA Bioresonan...

Anatomical image navigation

The interface allows the operator to:

- change between text and image lists;
- select anatomical sections;
- zoom into an area;

- open linked deeper images;
- choose cross-sections;
- add or remove measurement markers;
- draw a contour around a selected region.

These are mostly navigation operations over a library of anatomical illustrations, not automatically acquired CT, MRI or ultrasound images.

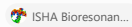
Nidus values

The software displays numbered markers commonly called **Nidus** markers. The manual describes a scale from 1 to 6:

1 → presented as better or more balanced
2
3
4
5
6 → presented as worse or more disturbed



The walkthrough says a value of four should be treated as serious within the software's own interpretation.



These values are not standard medical units. They are not equivalent to:

- blood concentrations;
- radiology measurements;
- electrical impedance in ohms;
- magnetic-field strength in tesla;
- laboratory reference intervals;
- validated disease probabilities.

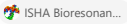
Etalon database

The software compares the selected result with stored reference entries, often called **etalons**.

Commonly advertised categories include:

Healthy organ models
Pathomorphology
Microorganisms
Biochemical homeostasis
Allergens
Food products
Herbs
Pharmaceutical products
Homeopathic preparations
Minerals and stones



The result screen may display a **coefficient of spectral similarity**, sometimes abbreviated CSS, and sort references according to the software's claimed similarity. The Hunter walkthrough explicitly describes pathology results being shown with CSS values. 

A likely logical structure is:

Etalon ID
Category
Name
Anatomical location
Reference curve or coefficient array
Display image
Interpretation text
Recommended related items
Language translations



But no public, authoritative technical specification identifies the numerical units, acquisition source or clinical reference process used to create each etalon.

What “25D software” claims to add

Third-party 25D sellers commonly claim that their software adds:

- more virtual organ and tissue models;
- cell, chromosome and DNA illustrations;
- additional pathology references;
- faster navigation into deeper image levels;
- new cancer-screening modules;
- additional therapy and comparison functions;
- more report languages.

These are **seller claims**, not independently established capabilities. One 25D product listing describes “25D” as a software module with additional virtual models and an oncology-related function, but supplies no controlled validation showing that the module can detect early cancer. 

The label therefore appears to describe the size and presentation of the software library—not a measurable hardware characteristic.

Meta Hunter versus 25D

Feature	Meta Hunter 4025	“25D” package
Controller	Hunter-style USB controller	Usually similar or identical-looking box
Headset	Paired earpiece sensors	Normally the same general headset design
Software	Metapathia GR Hunter-style UI	Modified or expanded Hunter-style software
Anatomy library	Organ and tissue images	Advertised as a larger image/database collection
Analysis	Nidus, etalons, CSS comparison	Same concepts with extra modules
True 25-dimensional sensing	Not documented	Not documented
Raw sensor export	Usually unavailable	Usually unavailable
Public calibration specification	Not found	Not found
Open AI model	Not available	Not available
Open etalon database	Not available	Not available
Clinical validity	Not established by the name	Not established by the 25D label

What to verify before purchasing one

Ask the seller for exact evidence, not screenshots or marketing descriptions:

1. Legal manufacturer name and address
2. Exact controller model and PCB revision
3. Device serial number
4. Software name, build number and release date
5. Whether the software is original IPP software
6. License and USB-dongle details
7. Complete user and service manuals
8. Raw sensor-data export capability
9. USB communication protocol
10. Sampling rate and ADC resolution
11. Headset sensor circuit and frequency response
12. Calibration fixture and calibration procedure
13. Etalon database schema and provenance



14. Regulatory certificate number and issuing authority
15. Clinical study matching the exact model and software build

A certificate logo printed on the enclosure is not sufficient. The certificate should identify the exact legal manufacturer, model, intended use and issuing body.

Relevance to your device project

For research, the most valuable parts of a Meta Hunter system would be:

- one complete licensed hardware unit;
- its exact software installation;
- the patient database format;
- the etalon files;
- USB traffic captures;
- headset electrical measurements;
- scans performed with known electronic test loads;
- scans with the headset disconnected;
- repeated scans without changing the patient;
- software output before and after modifying demographics.

That would help determine whether the controller transmits a meaningful measurement array or merely sends a small status/value packet that the software expands into reports.

The 25D label itself should not be used as the architecture for your new device. Your project should continue to store calibrated raw sensor measurements and train models against recognized reference tests rather than reproducing undocumented Nidus or etalon values.



so, what is the 21D, 25D related to ?

Thought for 10s >

In the Metatron/NLS market, **21D and 25D are mainly software-version or marketing labels**. They are not recognized scientific measurement units, and they do not mean the device physically measures 21 or 25 spatial dimensions.

For example, sellers describe **25D** as an expanded Metapathia software package with:

- more anatomical and tissue images;
- virtual views extending from organs to cells, chromosomes, and DNA;
- a larger “etalon” reference library;
- updated comparison algorithms;
- “multidimensional virtual scanning” and additional report modules.  eBay +1

So the intended meaning is roughly:

3D / 9D / 18D / 21D / 25D
↓
Seller-defined software generation or feature tier
↓
More images, database entries, analysis screens,
and claimed scanning depth



21D appears to be another reseller-defined release between 18D and 25D. Listings use it for systems with newer software, additional organ/cell libraries, or updated reports, but there is no consistent technical definition shared across vendors.  alibaba.co.uk +2

A simple comparison:

Label	Common marketplace meaning
3D	Basic anatomy visualization/software generation
9D	Expanded software and reference database
18D	More images, reports, and analysis modules
21D	Later reseller software edition
25D	Expanded "virtual scanning" library and additional modules

The number usually **does not describe**:

- the number of sensors;
- measured frequencies;
- ADC channels;
- magnetic-field dimensions;
- spatial imaging resolution;
- an AI-model architecture.

Also, different sellers may sell almost identical hardware under 18D, 21D, or 25D names. The actual difference may be only the installed software, database package, dongle license, language set, or branding.

For your own device, do not use labels like 21D or 25D as technical specifications. Use measurable descriptions such as:

16-frequency bioimpedance sweep
5 kHz–100 kHz
4-electrode measurement
16-bit ADC
complex impedance output
validated hydration model

📄

That tells engineers and researchers what the device truly measures.

📄👍🗣️🔄⋮🌐🔍📄

Sources

give me the url for the MetaTron 21D, 25D

Thought for 10s >

Here are the closest available product pages:

Meta Hunter / Metatron 21D

- [Alibaba — 21D-NLS 4025 listing/search page ↗](#)
This page includes a product listed as “21D-NLS 4025 9D 18D Comprehensive Body Detection System.”
I could not find a dedicated official manufacturer page defining a 21D model. 🇨🇳 Alibaba

Meta Hunter / Metatron 25D

- [Made-in-China — Quantum Resonance Magnetic 25D NLS ↗](#) 🇨🇳 Made-in-China
- [Alibaba — Metatron Hunter 4025 NLS 25D ↗](#) 🇨🇳 Alibaba
- [Global Sources — 25D Metapathia GR Hunter 4025 ↗](#) 🌐 Global Sources
- [SPW Industrial — Metatron Hunter 4025 25D ↗](#) 🇸🇬 SPW Industrial
- [eBay — Metapathia GR Hunter 4025 25D ↗](#) 🇺🇸 eBay

Official Metatron information

- [Institute of Practical Psychophysics — HSS Metatron device ↗](#) 🇩🇪 apparat-metatro...
- [How to recognize original Metatron packaging ↗](#) 🇪🇺 nls-metatron.eu

The 21D and 25D pages are reseller listings, and the labels are not consistently defined by the original Metatron manufacturer. Check the legal manufacturer, serial number, exact software build, licence dongle, and technical documentation before buying.

    ...  Sources

Mon, Jun 29 at 12:20 AM

I've heard the human health is related to the hurricain of the electromagnetic from sun

Thought for 28s >

You're probably referring to a **solar storm** or **geomagnetic storm**—"an electromagnetic hurricane from the Sun" is a reasonable informal picture, but it is not the scientific term.

The sequence is roughly:

Solar flare

→ burst of electromagnetic radiation

Coronal mass ejection

→ cloud of charged plasma and magnetic fields

Arrival near Earth

→ disturbance of Earth's magnetosphere

Geomagnetic storm

→ changes in Earth's magnetic field, auroras, induced currents, radio/GPS disturbances

NASA and NOAA explain that Earth's atmosphere and magnetic field protect people on the ground from most harmful solar radiation. The clearly established effects are mainly on satellites, radio communication, GPS, electrical grids, astronauts, and—in severe events—people flying at high altitude on polar routes.

 NASA Science +2

Does it affect human health?

There are **possible statistical associations**, especially involving:

- blood pressure;
- heart-rate variability;
- circadian rhythm and melatonin;
- sleep;
- cardiovascular events.

For example, a 2025 study analyzing more than 500,000 blood-pressure measurements found correlations between geomagnetic activity and population-level blood-pressure fluctuations. A 2022 cohort study reported reduced heart-rate variability during stronger geomagnetic disturbances.  Nature +1

However, the research is inconsistent. A 2020 replication study involving continuous heart-rate measurements found little or no reliable relationship after accounting for time-series effects. This means correlation has been observed in some studies, but **a direct biological cause has not been established**, and the effect—if real—is probably small compared with factors such as medication, sleep, temperature, stress, diet, exercise, age, and existing disease.  PubMed +1

So it would be too strong to say:

Human health is controlled by electromagnetic storms from the Sun.

A more accurate statement is:

Solar and geomagnetic activity may be a weak environmental factor affecting some physiological rhythms, but the magnitude, mechanism, and clinical importance remain uncertain.

Relevance to your device

You could include solar and geomagnetic activity as **research metadata**, but not use it as the primary health measurement.

For every device scan, store:

```
measurement_timestamp_utc
participant_location
Kp_index
Ap_index
Dst_index
F10_7_solar_flux
solar_wind_speed
interplanetary_magnetic_field_Bz
local_temperature
local_pressure
sleep_duration
blood_pressure
heart_rate
heart_rate_variability
```

Useful official sources are:

- NOAA’s **Kp index**, which characterizes the degree of geomagnetic disturbance; [NOAA Space We...](#) +1
- NASA’s **OMNIWeb**, containing Kp, Ap, Dst, solar-wind, magnetic-field and solar-flux data. [OMNIWeb Service](#) +1

Your research architecture could be:



Start by training the health model **without** solar variables. Then add Kp, Ap, Dst and solar-wind data and test whether performance improves on an independent participant-level test set. That would tell you whether solar activity contributes useful information instead of merely producing an interesting correlation.

... Sources

Mon, Jun 29 at 1:48 AM

so, for the metatron 21D / 25D, I'd like to know if it can increase the height of my daughter

Thought for 19s >

No—Metatron 21D/25D cannot reliably increase your daughter’s height. There is no credible clinical evidence that its headset, “frequency therapy,” etalon database, or bioresonance programs can lengthen bones, stimulate growth plates, or increase final adult height.

Height increases while cartilage growth plates are still open. After those plates close—usually near the end of puberty—even medically prescribed growth hormone does not make the long bones grow taller.

[MedlinePlus](#) +1

What matters is whether your daughter is:

- following her normal growth-chart curve;
- growing at an appropriate rate each year;
- entering puberty at an expected time;
- substantially shorter than expected from her parents’ heights.

A child below roughly the third height percentile, growing unusually slowly, or dropping across percentile lines should be assessed by a pediatrician. Evaluation may include accurate serial height measurements, parental heights, puberty assessment, a left-hand/wrist bone-age X-ray, and selected blood tests.

 HealthyChildren... +2

Real treatment depends on the cause. Most shorter children are healthy and do not need treatment. Growth hormone is reserved for particular diagnosed conditions and must be prescribed and monitored by a pediatric endocrinologist; it is not an ordinary height supplement.  MedlinePlus +1

Please do not let a Metatron scan persuade you that she has a hormone, vitamin, organ, or growth-plate problem, and do not give hormones or unverified supplements based on its report. The FDA has taken action against similar resonance-based devices promoted with unsupported claims about changing body functions or diagnosing disease.  U.S. Food and D... +1

Her age, current height, height one year ago, and both parents’ heights would allow a rough growth-pattern assessment—but her pediatrician’s measurements should guide any decision.

    ...  Sources

Mon, Jun 29 at 4:38 AM

is there any published design data for metatron 9D?

Thought for 28s >

Yes—but only **partial design information** is publicly available. I could not find a complete engineering package for a device specifically identified as **Metatron 9D**.

What is available

1. 9D-specific manuals and feature sheets

Public 9D-NLS manuals mainly describe installation, operation, anatomy screens, etalon comparisons, and claimed diagnostic functions. A published 9D feature sheet says the hardware connects through USB and contains:

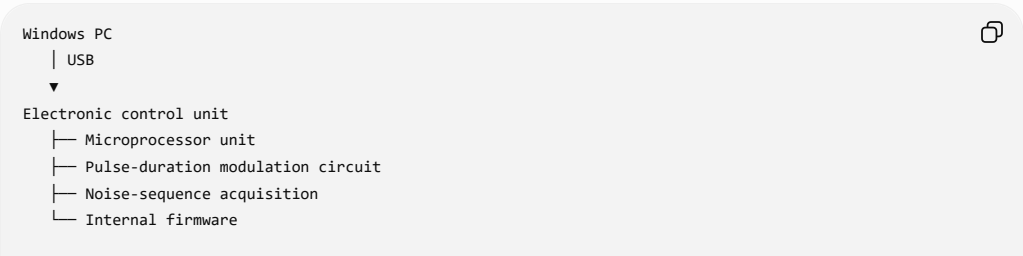
- a digital trigger sensor inside the controller;
- an analog trigger sensor inside the “bio-inductor” headset;
- Windows software and a USB licensing dongle.  Scribd +1

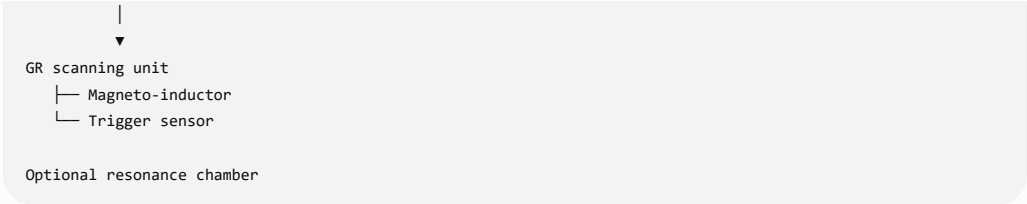
These documents do **not** include circuit schematics, component values, PCB files, sensor construction, firmware source, or a documented USB protocol.

2. Detailed older Metatron hardware specifications

The most detailed public technical document I found is the older **Metapathia-GR Hunter / Metatron operating manual**. It may describe the original Metatron architecture, but it cannot be assumed to match every reseller-branded “9D” device.

It lists this hardware structure:





The manual publishes these technical figures:

Parameter	Published value
Magnetic intensity at GR unit surface	21 ± 0.5 mT
Magneto-inductor interruption range	1.8–8.2 Hz
Frequency adjustment step	0.02 Hz
Low-frequency modulation	240 Hz
Claimed high-frequency modulation	4.9 GHz
Signal-processing band	4–1800 kHz
Differential amplifier gain	At least 85 dB
Shift-register clock	0.8 ± 0.01 MHz
Modulation method	Pulse-duration modulation
PC interface	USB
Software development environment	Borland Delphi 7
Database/storage	InterBase tables and binary files

It also describes the sensitive element as a noise-voltage generator using a modified **2G401B diode**, biased with several microamperes, with the resulting signal passing through an amplification path. [Altered States](#)

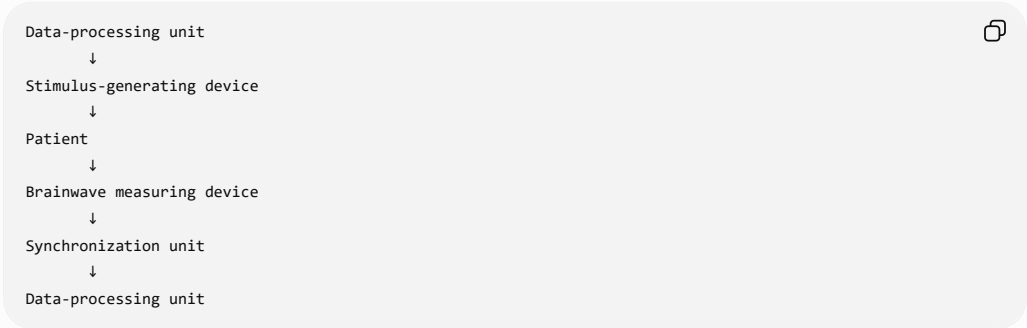
That is useful architectural information, but it is still not enough to reproduce the unit accurately.

Published patents

Two relevant patents provide functional diagrams and general system descriptions.

US 2010/0081959 A1 — Diagnostic Apparatus

This patent describes:



It also describes a stimulus-generating component containing a carbon single crystal positioned between electrodes. [Google Patents](#)

US 6,549,805 B1 — Torsion Diagnostic System

This earlier patent describes a central processing and telemetry unit, triggering sensors, operators, a patient, and a cavity resonator. It includes system-level figures and claims, but not a manufacturing schematic for a 9D controller.  Google Patents

The manufacturer lists both patents along with an industrial-design patent for the external apparatus.

 Metatron

What has not been published

I found no legitimate public release containing:

Complete circuit schematic

PCB layout or Gerber files

Bill of materials

Controller MCU model and pinout

Firmware source or binary specification

USB VID/PID and command protocol

ADC type and sampling format

Headset coil winding data

Trigger-sensor circuit

Microwave/RF circuit design

Calibration fixture and procedure

Raw scan-data format

Etalon database schema

Numerical etalon records

Health-analysis algorithms or model weights

Clinical reference dataset



So the available material is enough to create a **conceptual block diagram**, but not enough to clone or independently verify the exact Metatron 9D.

Important problem with the “9D” name

“9D” does not appear to identify one standardized hardware revision. Marketplace devices sold as 8D, 9D, 17D, 18D, or 25D may use different controller boards, software, databases, dongles, and headset electronics despite having similar enclosures. Current listings frequently combine several “D” versions in one product description, which makes the label unreliable as an engineering identifier.  Alibaba +1

To identify the design of one particular unit, you would need at minimum:

Clear exterior photographs

Rear-panel and connector photographs

Manufacturer and model label

Software name and version

USB VID and PID

PCB photographs of both sides

IC package markings

Headset internal photographs

USB traffic captured during one scan



The public documents show the claimed architecture, but a physical teardown and protocol capture would be necessary to establish what a particular 9D unit actually measures.

    ...  Sources

can you get the AI model for the 9D ?

Thought for 23s >

I could not find a legitimate public download of an **AI model for “Metatron 9D.”** Based on the available documentation, it may not use a modern neural-network model at all.

Public Metatron materials describe a system that:


```

Measured or generated signal
↓
Proprietary processing algorithm
↓
Comparison with stored reference "etalons"
↓
Spectral-similarity coefficients
↓
Anatomical display and report

```



The manufacturer says its current software contains more than **12,000 reference etalons** and that the database and algorithms are continuously updated. However, it does not publicly provide model weights, training records, numerical etalon arrays, source code, an API, or an SDK. Metatron +2

The older Metapathia-GR manual also describes "algorithms introduced into the program," rather than a separately downloadable AI model. Altered States

What "the model" probably consists of

It is likely a proprietary package containing several parts:

1. Reference etalon database
2. Spectral-similarity calculation
3. Entropy/NLS analysis rules
4. Anatomical image database
5. Report-generation rules
6. Patient database
7. Hardware communication driver
8. License or USB-dongle protection



That is closer to a **reference-matching and rules engine** than a contemporary trained AI model such as a neural network.

The "9D" problem

I could not confirm "9D" as a standardized official model name on the original manufacturer's current pages. Official materials list systems such as Metatron-4025 and software editions such as Metapathia Hospital, WellBeing, Sakura, Prana, and Intruder. "9D" is commonly used by resellers, so different 9D packages may contain different hardware, databases, or modified software. Metatron

Legitimate ways to obtain it

1. Request licensed technical access

Contact the Institute of Practical Psychophysics and ask for:

- Research or developer licence
- Metatron-4025-compatible software
- Etalon database licence
- Raw scan-data export
- Database data dictionary
- Device communication protocol
- SDK or API availability
- Sample anonymized examinations
- Algorithm documentation
- Calibration documentation



The manufacturer says it provides remote technical support and software updates to customers. Its published contact email is ipp.info@bk.ru. Metatron +1

2. Inspect a licensed installation

When you own a licensed system, I can help determine whether it contains:

```

.db
.sqlite
.mdb
.dat

```

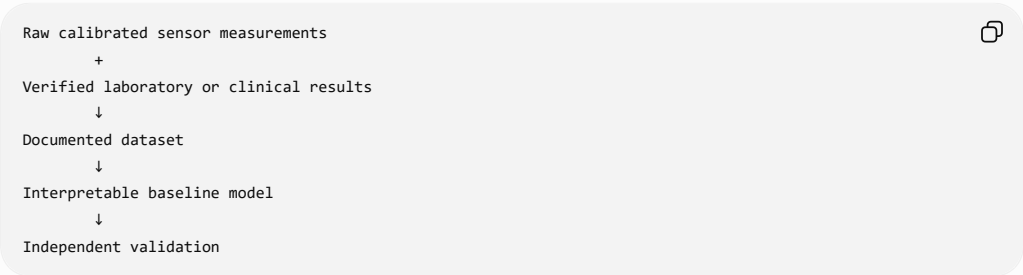


```
.bin
.xml
.json
.csv
```

We can inventory and analyze accessible files and exported examinations without bypassing the activation dongle or copying protected software.

3. Build an independent model

For your own device, the scientifically stronger route is:



Do not train your model using Metatron reports as the medical truth. That would reproduce Metatron-style results without establishing that they correspond to real health measurements.

So: I cannot currently retrieve a public Metatron 9D AI model because none appears to have been released. The practical next step is obtaining licensed software or a research agreement from the developer, then examining its permitted exports and database structure.

📄 👍 🗨️ ↺ ⋮ 🌐 Sources

Mon, Jun 29 at 6:17 AM

then can you find the database or experiment data for the 9D

Thought for 29s >

I searched specifically for **Metatron/9D NLS etalon databases, patient datasets, raw scan records, and experimental data**. I did **not** find a legitimate public download containing the numerical 9D database or raw device measurements.

What I found

1. Official Metatron reference database

The original Metatron developer says its current systems contain more than **12,000 reference “etalons.”** The listed groups include:

- virtual organ models;
- biochemical homeostasis;
- diseases or “nosology”;
- bacteria, viruses, fungi and parasites;
- allergens;
- medicines, supplements and foods;
- trace elements and psycho-emotional categories.

However, the official page only describes the categories. It does not provide CSV, SQL, JSON, spectra, model weights, record definitions, or downloadable numerical etalons.  Metatron

This means the useful database appears to be bundled inside licensed software rather than published as a research dataset.

2. Metapathia patient and examination database

The public **Metapathia GR Hospital manual** shows that the software locally stores:

- patient demographics;
- examination dates;
- selected organ images;
- resonance-frequency graphs;
- previous examinations;
- laboratory blood and urine results entered by the operator;
- generated reports.

The manual says reports can be exported as PDF, but it does not describe an export of raw sensor samples or numerical etalon curves. It also states that the patient card displays an organ image and that selecting it reveals the organ-modulation frequency graph. imda.moh.gov.vn

So there is clearly a local database, but the published manual does not reveal its internal file schema.

3. Published experimental material

I found one peer-reviewed case report using **Metatron Hunter** during ozone treatment for one patient with fibromyalgia. Scans were reportedly performed before, during and after treatment. The paper provides a small summary table of percentage changes, but no raw spectrum, no individual frequency samples, no database export and no downloadable patient file. [Semantic Scholar](#)

This is therefore **not suitable as an AI training dataset**.

The manufacturer's brochure also advertises diagnostic accuracy figures such as 89.8% for gastroenterology and lower values for other areas, but the brochure does not include participant-level records, confusion matrices, raw scans, train/test splits or enough methodology to reconstruct those results. [Pub MediaBox S...](#)

4. Articles describing NLS experiments

The manufacturer-associated site hosts papers concerning topics such as digestive organs, lymph nodes, pulmonary aspergillosis, pyelonephritis, adrenal tumours and liver conditions. These are article PDFs rather than downloadable datasets, and I found no accompanying raw-data repositories. [uk.metatron-nls... +4](#)

What is not publicly available

I could not find any legitimate public source containing:

9D raw sensor recordings
Numerical etalon frequency arrays
Etalon database SQL/CSV files
Patient examination files
Device calibration records
USB protocol documentation
Sampling frequency and ADC output
AI training records
AI model weights
Clinical ground-truth labels
Data dictionary for 9D reports



Also, "9D" appears primarily in reseller documentation, not as a clearly standardized data format from the original developer. Different 9D systems may contain different software builds and databases.

Best way to obtain usable data

Contact the Institute of Practical Psychophysics and request a **research or developer-data licence**. Its official software page provides the contact address ipp.info@bk.ru. [Метатрон](#)

Ask for these exact materials:

1. Numerical etalon database licence
2. Etalon record data dictionary
3. Raw examination export format
4. Anonymized sample patient examinations
5. Raw sensor signal or frequency-array export
6. Device-to-PC communication protocol
7. Calibration and reference-load procedure
8. SDK or API documentation



- 9. Clinical-validation datasets
- 10. Exact software and hardware versions used in each study

A useful email wording is:

We are developing a research system and are seeking licensed access to Metatron/NLS data. Please advise whether you provide a research licence for the numerical etalon database, anonymized examination records, raw sensor output, calibration documentation, file-format specifications, API/SDK access and the clinical reference datasets used to validate the system.

When you have a licensed 9D installation

We can inspect the installation to identify accessible database files without bypassing licensing or copying protected software.

Run this PowerShell inventory:

```
Get-ChildItem "C:\Path\To\9D" -Recurse -File |
    Group-Object Extension |
```

<> PowerShell

Then search likely database formats:

```
Get-ChildItem "C:\Path\To\9D" -Recurse -File |
    Where-Object {
        $_.Extension -in @(
            ".db", ".db3", ".sqlite", ".sqlite3",
            ".mdb", ".accdb", ".fdb", ".gdb",
            ".dat", ".bin", ".xml", ".json",
            ".csv", ".ini"
        )
    }
```

<> PowerShell

The older Metatron materials mention InterBase-style storage, so .gdb or .fdb files would be especially interesting, although a particular reseller's 9D software may use something completely different.

Realistic conclusion

There are published descriptions and small case studies, but no open 9D training database comparable to a normal medical-AI dataset. The most valuable next step is obtaining one licensed installation and determining:

- What data the controller actually sends
- What numerical data the software stores
- Whether scans can be exported
- Whether etalons contain real arrays or only packaged rules
- Whether results change with known electronic test inputs

A copy of the 9D installation-folder listing, exported examination, or sample report would let me identify the likely database structure and design an extraction plan.

📄 👍 💬 ↺ ... 🌐 📄 Sources

I'd like to know the mora, zipper devices in more detail

Thought for 40s >

MORA and Zapper devices are different types of alternative-health equipment, although sellers often mix words such as “frequency,” “resonance,” and “energy.”

1. MORA bioresonance device

MORA comes from the names of physician Franz **Morell** and engineer Erich **Rasche**, who developed the method in the 1970s. It grew out of electrodermal testing and is marketed for both assessment and treatment. eprints.aihta.at

Marketed operating principle

The claimed workflow is:



MORA proponents claim that electrodes receive the body’s electromagnetic “oscillations,” that the machine separates “healthy” and “pathological” components, and that undesirable components can be inverted or modified before being returned through another electrode. The health-technology assessment that described this process also noted that the proposed physical and physiological mechanisms remain hypothetical. eprints.aihta.at

Typical hardware

A modern MORA system may contain:

- hand, foot, head, back or point electrodes;
- separate input and output channels;
- high-gain analog amplifiers;
- selectable filters;
- phase-inversion or non-inversion modes;
- square- and sine-wave generators;
- an integrated touchscreen computer;
- input and output cups for ampoules, foods or supplements;
- stored programs for allergies, acupuncture points and other advertised applications.

One MORA Nova product page lists two channels, a claimed operating range of **0.1 Hz to 1 MHz**, adjustable filtering, amplification, and internal sine- and square-wave generators. Those are manufacturer specifications; they do not demonstrate that the signals represent organs, diseases or allergens. [Mora Turkey](https://mora-turkey.com)

The cup or ampoule system

Some MORA workflows place an allergen, food, medicine or ampoule into an input cup. The software is then claimed to introduce that substance’s “frequency information” into the processing path.

The same concept may also be supplied digitally as “electronic homeopathy.” The public literature does not show a reproducible method proving that a material in a metal cup produces a unique clinical signal capable of diagnosing intolerance or predicting whether a medicine will work. [eprints.aihta.at](#)

Does MORA genuinely measure health?

The key engineering problem is that public material generally does not disclose:

Raw waveform units
Sensor noise floor
Input impedance
Calibration reference
Signal-to-noise ratio
Biological source of each frequency
Algorithm for “healthy/pathological” separation
Clinical labels used to create programs
Validated diagnostic thresholds



Without these, its output cannot be treated like ECG, EEG, bioimpedance spectroscopy, MRI or a laboratory assay.

Clinical evidence

The evidence is mixed but generally weak:

- A double-blind trial involving 36 children with atopic dermatitis found improvement in both the real and placebo groups, with **no significant difference between them**. [eprints.aihta.at](#)
- A small gastrointestinal study involving about 21 patients reported favorable results, but the health-technology review rated the strength of evidence as low because of the small number of participants and large variation. [eprints.aihta.at](#)
- A 190-person pilot study reported better smoking-cessation results than placebo, but one pilot study does not validate MORA for unrelated diseases, allergies, hormone problems or growth. [Karger Publishers](#)
- As of January 2026, the UK advertising regulator stated that it had not seen robust clinical evidence supporting bioresonance devices for diagnosing, preventing or treating diseases. [ASA](#)

So MORA should be regarded as an **unvalidated complementary therapy**, not a general diagnostic instrument.

2. Zapper device

A **Zapper** normally does not analyze a returning signal. It simply applies a small electrical waveform through skin electrodes or metal handholds.

Basic architecture

Battery
|
▼
Oscillator
|
▼
Square-wave generator
|
▼
Output resistor/current limiting
|
▼
Two handholds or skin electrodes
|
▼
Current passes through part of the body



The classic device associated with Hulda Clark is commonly described as applying a positively offset square-wave signal around 30 kHz, although commercial “zappers” now use many different frequencies and

waveforms. A later patent discussing that design also notes the positive offset and 30 kHz configuration.

 Google Patents

What it claims to do

Zapper sellers have claimed that selected frequencies can:

- kill parasites;
- eliminate bacteria or viruses;
- destroy fungi;
- treat cancer;
- improve immune function;
- “detoxify” the body;
- correct organ frequencies.

The underlying idea is often that every pathogen has a vulnerable resonant frequency and can be destroyed electrically without damaging human tissue.

There is no accepted clinical evidence that a small current applied through handhelds can selectively kill pathogens inside the body. The FTC brought action over unsupported claims that a “Super-Zapper” could kill bacteria, viruses and parasites and cure cancer, AIDS, diabetes and other serious diseases.  Federal Trade C...

Does a zapper contain a database or AI?

Usually, no. A simple zapper contains only:

Fixed oscillator
or
Selectable frequency generator
+
Timer
+
Output stage



More expensive models may include a “frequency library” with labels such as:

Parasite program
Virus program
Liver program
Cancer frequency
Detox program



Those are normally preset frequency tables—not clinically trained AI models or validated pathogen databases.

3. MORA versus Zapper

Feature	MORA	Zapper
Reads a signal	Claims to receive a body signal	Usually no
Sends a signal	Yes	Yes
Main interface	Multiple electrodes and cups	Handholds or adhesive electrodes
Processing	Filtering, amplification and inversion	Waveform generation
Typical software	Patient records and therapy programs	Timer and frequency presets
Claimed use	Assessment and treatment	Killing pathogens or stimulation
Open clinical database	No	No

Feature	MORA	Zapper
Open AI model	No	Usually no AI at all
Proven disease diagnosis	No	No
Proven height increase	No	No

4. How they differ from legitimate electrical stimulation

Electrical stimulation itself is not imaginary. Legitimate devices include:

- TENS for certain pain-relief applications;
- powered muscle stimulators;
- implanted cardiac stimulators;
- peripheral nerve stimulators;
- approved neurostimulation systems.

Those devices use defined waveforms, electrode placements and specific medical indications. They do **not** claim that one frequency diagnoses every organ or kills every pathogen.

The FDA recognizes powered muscle stimulators for limited uses such as muscle re-education, relaxation of spasms and maintaining range of motion—not for curing infections or universal disease treatment.

 U.S. Food and D...

5. Electrical safety

A zapper or MORA-like output is not automatically harmless merely because it uses a battery.

Electrical stimulation should not be applied:

- across the chest;
- over the neck or carotid sinus;
- through the head;
- near implanted pacemakers or other electronic implants;
- over irritated, infected or damaged skin.

FDA guidance warns that transthoracic current can cause arrhythmias, neck stimulation can interfere with breathing, and skin irritation can occur. Safety during pregnancy has not been established for powered stimulators.  U.S. Food and D...

For a child, especially your daughter, I would not use either device for growth stimulation. **Neither MORA nor a zapper has a mechanism or reliable evidence for increasing bone length, reopening growth plates or increasing final height.**

6. For your research project

The useful engineering lesson is:

MORA

= electrode feedback-processing concept

Zapper

= electrical waveform generator

Your proposed V0.1

= calibrated impedance measurement instrument



Your V0.1 architecture is scientifically stronger because it produces measurable quantities:

Frequency in hertz

Resistance in ohms

Reactance in ohms

Phase in degrees

Calibration error

Signal-quality score



MORA’s “pathological frequency,” a zapper’s “parasite frequency,” and Metatron’s “etalon similarity” are not equivalent to those calibrated physical measurements.

📄 👍 💬 ↺ ⋮ 📄🌐🔍 Sources

which one is better?

Thought for 9s >

For health measurement, disease treatment, or increasing your daughter’s height, neither MORA nor a Zapper is a good choice.

Goal	Better option
Measure health metrics	Your calibrated multi-frequency bioimpedance prototype
Diagnose illness	Medical examination and validated laboratory/imaging tests
Treat infection, TB, cancer, or parasites	Evidence-based medical treatment
Increase a child’s height	Pediatric/endocrinology assessment
Electrical pain relief	A properly cleared TENS device, when medically appropriate
Research frequency response	Laboratory signal generator and calibrated impedance instrumentation

MORA

MORA is more elaborate: it claims to receive body signals, filter or invert them, and return modified signals. However, regulators state that robust clinical evidence is lacking for bioresonance diagnosis and treatment claims. 🇺🇸 ASA +1

Zapper

A Zapper is basically a waveform generator connected to electrodes or hand grips. Claims that it kills parasites, viruses, bacteria, or treats serious diseases have been subject to FTC enforcement as unsubstantiated. 🇺🇸 FTC Federal Trade C... +1

It also directly applies current, so misuse can cause shocks, burns, skin irritation, pain, or interference with implanted electronic devices. Electrical stimulation across the chest or head can be dangerous.

🇺🇸 U.S. Food and D... +1

My recommendation

For the device you are designing:

Best: calibrated four-electrode bioimpedance analyzer

↓

Measure resistance, reactance and phase

↓

Validate against known R/C loads

↓

Collect paired clinical-reference data

↓

Build one validated model

📄

Do not copy the MORA or Zapper architecture as a medical-treatment system. Your **AD5940-based V0.1 measurement device is the better scientific foundation** because it produces measurable physical values in ohms and degrees rather than undocumented “pathological frequencies.”

So, between the two: **MORA is more sophisticated-looking, but neither is clinically preferable. A validated bioimpedance instrument or approved medical stimulator is the better route.**

       Sources

what is difference from Metatron 9D?

Thought for 17s >

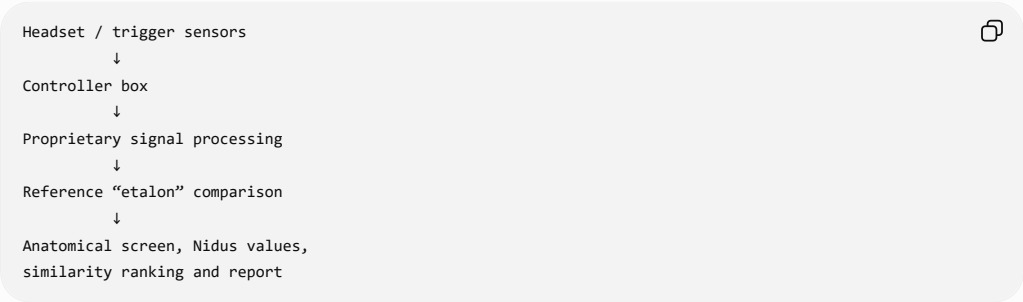
Metatron 9D is mainly a software-driven “scan and compare” system, while MORA is a bioresonance feedback device and a Zapper is mostly a signal generator.

Main difference

Device	Main function claimed	Typical hardware	Data processing
Metatron 9D / NLS	Analyze organs and tissues by comparing signals with stored references	Headset/“trigger sensors,” controller box, Windows PC	Compares results with a proprietary “etalon” database and displays anatomical images and similarity scores
MORA	Receive, modify, and return alleged body electromagnetic signals	Hand/foot electrodes, input and output channels	Filters, amplifies, passes, or inverts signals before returning them through electrodes
Zapper	Apply an electrical waveform to the body	Oscillator, current-limiting output, hand grips or pads	Usually generates fixed or selectable frequencies; little or no analysis

Metatron 9D

A typical claimed workflow is:



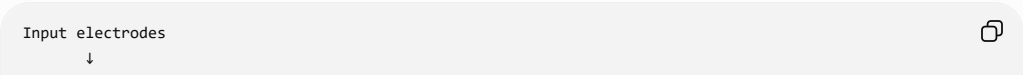
The original Metatron developer describes software containing more than **12,000 reference etalons**, grouped into areas such as anatomical models, biochemical homeostasis, microorganisms, allergens, medicines, and foods. The underlying numerical arrays, calibration method, raw sensor format, and model weights are not publicly disclosed.  Metatron +1

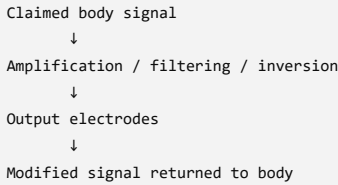
The label “9D” is **not a recognized physical dimension or standardized hardware specification**. Reseller pages mix labels such as 8D, 9D, 18D, and 25D with similar-looking hardware, suggesting that the number often describes a software tier, database edition, or marketing generation rather than a measurable sensor property.  Alibaba +1

Some reseller versions add a function called **Meta-Therapy**, which claims to send corrective frequencies. That gives Metatron some overlap with MORA or a programmable Zapper, but its primary interface remains the anatomical and etalon-comparison software.  Gayatri AI

MORA

MORA is more directly a **feedback-processing device**:



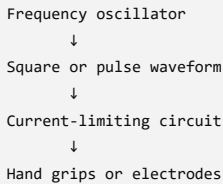


The manufacturer describes body signals entering through electrodes, being processed according to selected settings, and then being fed back through output electrodes. [moratechnologi...](#)

Unlike Metatron, MORA normally does not center its workflow around thousands of anatomical images and etalon rankings. Its distinguishing feature is the claimed processing and return of endogenous signals.

Zapper

A Zapper is much simpler:



It generally does not measure an organ signal, create anatomy images, or compare a scan against a database. Its “programs” are normally just preset frequencies and durations.

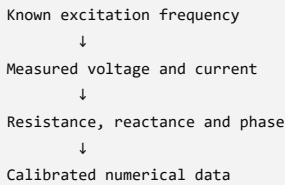
The FTC has taken action against marketers that claimed Zappers could kill bacteria, viruses, or parasites and treat serious diseases because those claims were not adequately substantiated. [FTC Federal Trade C... +2](#)

Simple analogy

Metatron 9D	= scanner-like software + proprietary reference library
MORA	= claimed signal feedback and inversion processor
Zapper	= electrical frequency generator

Compared with your V0.1 device

Your proposed AD5940 design is fundamentally different:



It produces traceable physical units such as ohms and degrees. Metatron’s Nidus values, MORA’s “pathological oscillations,” and Zapper frequency labels are not equivalent to calibrated bioimpedance measurements.

There is currently no robust evidence that bioresonance devices can broadly diagnose or treat medical diseases, according to the UK advertising regulator’s evidence review. [asa.org.uk](#)

... Sources

what is common or similar between three devices?

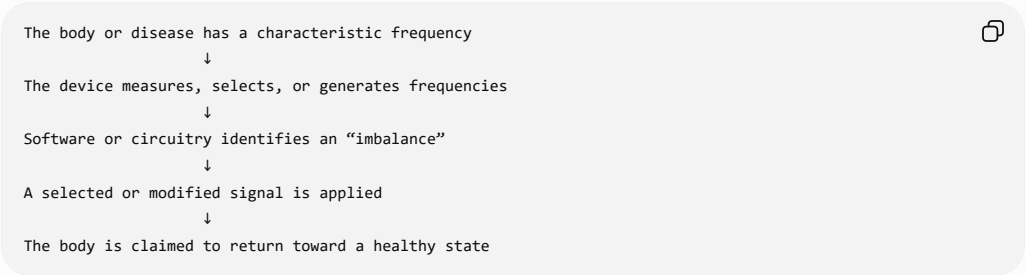
Thought for 9s >

The three devices—**Metatron 9D**, **MORA**, and **Zapper**—share the same broad “frequency medicine” **concept**, but implement it differently.

Common feature	Metatron 9D	MORA	Zapper
Uses electrical/electromagnetic terminology	Yes	Yes	Yes
Contacts the user through a headset, electrodes, or hand grips	Headset	Electrodes	Hand grips/electrodes
Uses selected frequencies or waveforms	Claimed	Yes	Yes
Has preset programs or reference information	Large “etalon” database	Therapy programs/filter settings	Frequency lists
May deliver a signal back to the user	Some models/Meta-Therapy	Yes	Yes
Claims to influence health	Yes	Yes	Yes
Publicly documented clinical AI model	No	No	Usually no AI
Robust evidence for broad diagnosis/treatment	Not established	Not established	Not established

Their shared idea

All three are marketed around roughly this concept:



The central problem is that vendors generally do not publish reproducible evidence establishing:

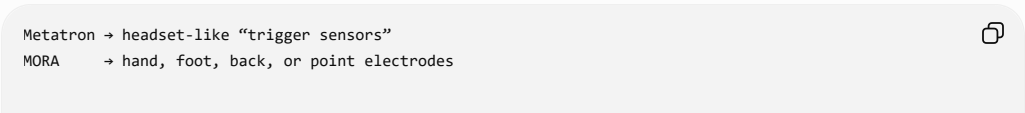
- which exact physiological source produces each frequency;
- how the signal is calibrated;
- how an organ or disease is distinguished from noise;
- how frequency libraries were created;
- comparison against accepted laboratory or imaging tests;
- raw patient datasets and independently validated algorithms.

The UK Advertising Standards Authority describes bioresonance as claiming to read energy wavelengths from cells, but says it has not seen robust clinical evidence supporting disease diagnosis or treatment with such devices. 

Similar hardware concepts

1. A user-contact interface

Each device connects to the user:



Zapper → metal hand grips or adhesive electrodes

2. Signal-generation electronics

They usually contain some combination of:

- Oscillator
- Frequency generator
- Amplifier
- Filter
- Output driver
- Timer
- USB interface



A Zapper usually uses the simplest architecture. MORA adds input processing and output feedback. Metatron adds a computer interface and a large proprietary reference library.

3. Frequency/program libraries

They all may associate preset signals with labels such as:

- Liver
- Allergy
- Parasite
- Virus
- Detox
- Stress
- Immune system



But these labels do not automatically mean the frequencies were validated as unique measurements of those conditions.

Metatron's developer says its software contains more than 12,000 reference "etalons," covering anatomical models, microorganisms, allergens, foods, and other categories. The underlying numerical data and validation dataset are not publicly released. [Metatron](#)

4. Proprietary interpretation

All three typically provide conclusions without making the full algorithm publicly auditable:

```

Input or selected frequency
↓
Proprietary processing
↓
Named program, similarity value,
imbalance, or treatment recommendation
  
```



The most important difference

Their primary roles are different:

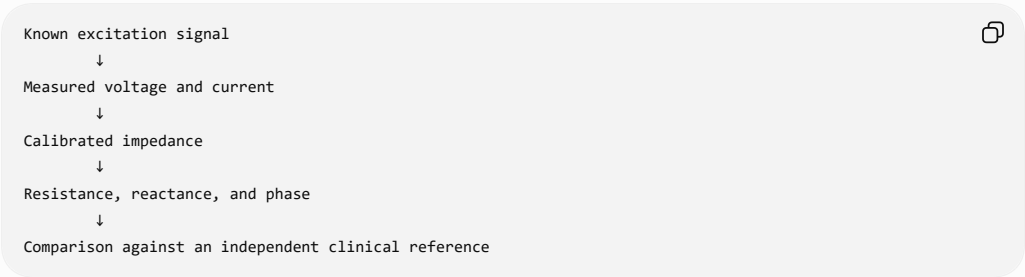
Metatron 9D	
= claimed scanner + anatomical software + etalon comparison	
MORA	
= claimed body-signal input + filtering/inversion + feedback output	
Zapper	
= waveform generator applying preset electrical signals	

MORA-related designs explicitly describe receiving signals through an electrode, processing them, and returning output signals through another electrode. [Google Patents](#)

A typical Zapper does not perform meaningful analysis—it mostly generates a waveform. The FTC has acted against sellers making unsupported claims that Zappers could kill bacteria, viruses, and parasites or treat serious illnesses. [FTC Federal Trade C...](#) +1

Compared with your V0.1 device

Your AD5940 bioimpedance project uses frequencies too, but the scientific architecture is different:



That produces measurable physical units. Metatron’s etalon similarity, MORA’s “pathological oscillation,” and a Zapper’s disease-named frequency program are not equivalent to calibrated impedance measurements.

...

Sources

give me the url for each device

Thought for 9s >

Here are the main pages for each device:

- **Metatron NLS:** [Official HSS Metatron device page ↗](#) — from the Institute of Practical Psychophysics. I could not verify an official manufacturer model specifically named “**Metatron 9D**”; 9D is mostly used by third-party resellers. 🌐 Metatron
- **MORA:** [Official MORA Nova page ↗](#)
Alternative manufacturer page: [Med-Tronik MORA Nova ↗](#) 🌐 moratechnologi... +1
- **Zapper:** [Dr. Clark equipment and Zapper products ↗](#)
“Zapper” is a generic device category, so there is no single official manufacturer or standardized model. This store sells a representative Clark-style Zapper. 🌐 drclarkstore.com

For context, the FTC has previously challenged unsupported disease-treatment claims made for Clark-style Zappers: [FTC case information ↗](#). 🇺🇸 ftc.gov

...

Sources

how much is the price of each device?

Thought for 33s >

Approximate **current market prices as of June 2026:**

Device	Typical price
Generic “Metatron 9D/25D” reseller unit	\$300–\$900
Meta Hunter / Metatron 4025 package	\$1,300–\$4,600
Original HSS Metatron	Quote required
MORA Nova	Roughly €10,000–€15,000+; used examples around \$13,650
Basic Clark-style Zapper	\$35–\$150
Branded SyncroZap	About \$290–\$300

Device	Typical price
Advanced programmable Zapper	\$300–\$1,200

Metatron

Low-cost reseller listings currently include a generic 9D unit around **\$294**, an Alibaba Meta Hunter sample around **\$850**, and another Hunter 4025 listing around **\$1,362**. 

More complete Hunter 4025 bundles with a laptop, software, dongle, support, and accessories are listed around **\$3,963–\$4,565**.  +1

The original Institute of Practical Psychophysics does not publish a price on its purchase page; buyers must contact the company or a representative for a quotation. 

MORA Nova

MORA Nova is modular, and the official seller prepares an individual quotation depending on the installed modules. 

A publicly archived used MORA Nova listing was approximately **\$13,650**, excluding shipping and VAT. A third-party market estimate places new systems around **€10,000–€15,000 or more**, depending on configuration and training.  +1

Zapper

Simple Clark-style units commonly range from about **\$35 to \$150**. 

The Dr. Clark Store lists its SyncroZap A11 at **\$289.99 with wristbands** or **\$299.99 with copper tubes**. 

More elaborate programmable models range from roughly **\$300 to \$1,200**.

The big price differences usually reflect software, database licences, laptop, accessories, training, branding, and reseller markup—not necessarily different sensing hardware or better medical accuracy.

 Sources

how about PEMF device?

Thought for 34s >

PEMF—pulsed electromagnetic field—devices are more scientifically grounded than Metatron, MORA, or Zapper systems, but only for narrow, device-specific uses. They generate timed magnetic pulses through a coil; they do not scan organs, read “resonance diseases,” or use an etalon database.

Typical prices

Type	Approximate current price
Small local applicator	\$170–\$400
Consumer pad or chair mat	\$300–\$900
Full-body home mat	\$700–\$2,000
Advanced home system	\$3,000–\$8,000
Professional high-intensity system	\$8,000–\$20,000+

Type

Approximate current price

Prescription bone-growth stimulator

Usually physician/insurance supplied; cash pricing is often not publicly listed

OMI PEMF Medallion
\$171.00 • Compact PEMF medallion for targeted therapy.

OMI Pulsepad
\$179.55 • Portable pad ideal for on-the-go relief.

NeoRhythm OmniPEMF
\$349.00 • Wearable PEMF device with versatile use.
★ 4.3

What PEMF can legitimately do

FDA-approved PEMF bone-growth stimulators exist for specific situations such as established fracture nonunion, failed spinal fusion, or improving fusion probability after certain spinal-fusion surgeries. For example, CervicalStim is indicated as an adjunct after cervical fusion in high-risk patients—not as a general wellness or growth device. [FDA Access Data +1](#)

Evidence for broader uses such as osteoarthritis, back pain, and musculoskeletal pain is mixed. Some reviews report modest improvements in function or pain, but protocols vary considerably and stronger trials are still needed. A 2024 systematic review did not support PEMF for improving healing of ordinary acute fractures. [PubMed +2](#)

Compared with the other devices

Device	Primary function
Metatron 9D	Claims to scan and compare results with a proprietary etalon database
MORA	Claims to receive, modify, and return body signals
Zapper	Sends electrical pulses through electrodes
PEMF	Creates a magnetic field around a treatment area using an energized coil

A properly designed PEMF system can have defined parameters:

Pulse frequency: Hz
Magnetic flux density: μ T or mT
Pulse width: μ s or ms
Waveform
Coil geometry
Duty cycle
Treatment duration

That makes it easier to reproduce and scientifically test than undocumented “9D” or organ-frequency claims.

Important buying warning

A seller saying “FDA registered” or “FDA listed” does not mean the device has been reviewed, cleared, or approved for its advertised health claims. FDA explicitly says registration and listing do not denote approval or authorization. [FDA Access Data +1](#)

Look for the exact:

Manufacturer

Model number

FDA PMA or 510(k) number

Approved/cleared indication

Magnetic intensity

Frequency range

Pulse waveform

Treatment protocol

Clinical evidence for that exact model

Regarding your daughter’s height

A PEMF device should **not** be purchased to increase her height. Bone-growth stimulators help certain fractures or surgical fusions heal; they do not lengthen normal leg or spine bones, reopen growth plates, or increase final adult height. The right next step remains a pediatric growth evaluation, especially because you previously mentioned no measurable growth for six months.

       Sources

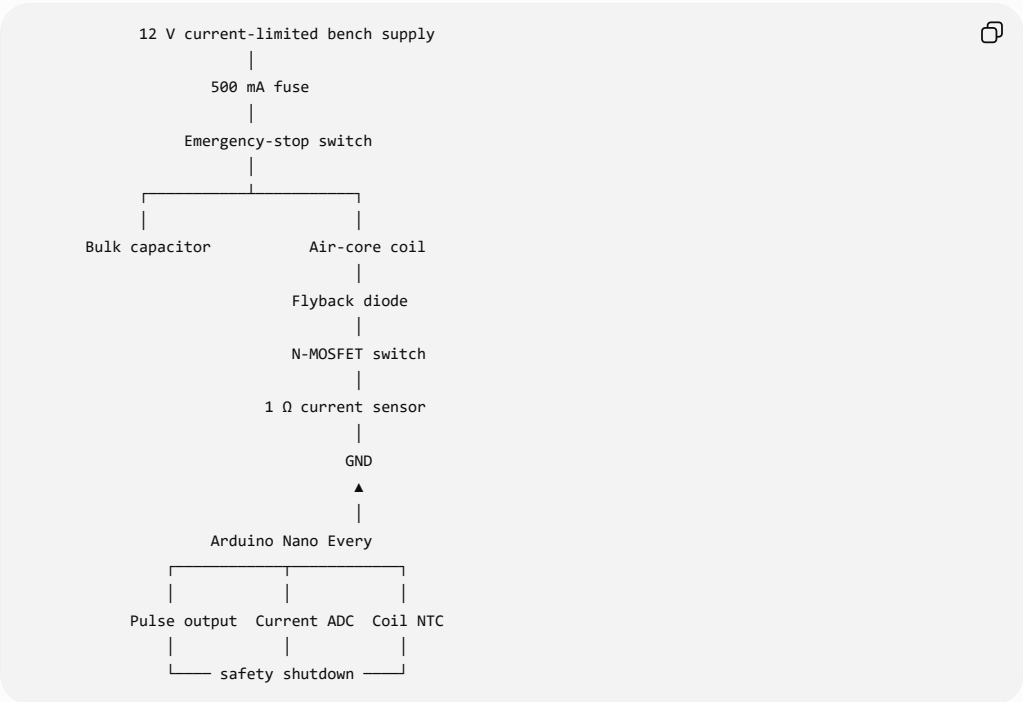
can you give me full design of a simple PEMF device?

Thought for 33s >

I can give you a **complete bench-test PEMF demonstrator**, but not a patient-use medical-device design. A device intended for treatment requires independently engineered current limits, thermal protection, EMC testing, risk management, and clinical/regulatory validation. FDA-authorized PEMF devices are approved for specific indications such as certain bone-healing applications—not general wellness, height growth, infection treatment, or organ “frequency correction.”  U.S. Food and D... +1

[Download the PEMF Bench V0.1 design package](#)

Block diagram



The Arduino Nano Every operates at 5 V and provides PWM-capable digital pins; the package uses D9 as the MOSFET-control output.  Arduino Docum... +2

Default bench configuration

Supply:12 V DC

Supply current limit:0.35 A

Pulse frequency:10 Hz

Pulse width:500 μs

Maximum frequency:50 Hz

Maximum pulse width:1,000 μs

Maximum session:60 seconds

Current shutdown:0.30 A

Temperature shutdown:40 °C

These are conservative **prototype limits**, not therapeutic settings.

Main components

Component	Suggested part
Controller	Arduino Nano Every
MOSFET	IRLZ44N logic-level N-channel
Flyback diode	SS34, 3 A/40 V
Current sensor	1 Ω, 1 W resistor
Gate resistor	100 Ω
Gate pulldown	100 kΩ
Temperature sensor	10 kΩ NTC, B≈3950
Input fuse	500 mA
Bulk capacitor	470 μF/25 V
Coil	500 turns AWG30, approximately 100 mm diameter
Power source	Current-limited laboratory supply

The IRLZ44N is a 55 V logic-level MOSFET with specified low on-resistance at a 4.5 V gate drive, making it practical for a 5 V microcontroller-controlled bench switch. [Infineon +1](#)

The flyback diode is essential because the coil is an inductive load. When the MOSFET turns off, the diode provides a controlled path for the coil current and limits the voltage transient that could otherwise damage the switch. [Texas Instruments +1](#)

Coil specification

Former diameter:100 mm

Turns:500

Wire:AWG30 enamelled copper

Winding width:approximately 20 mm

Estimated resistance:45–65 Ω

Estimated inductance:20–50 mH

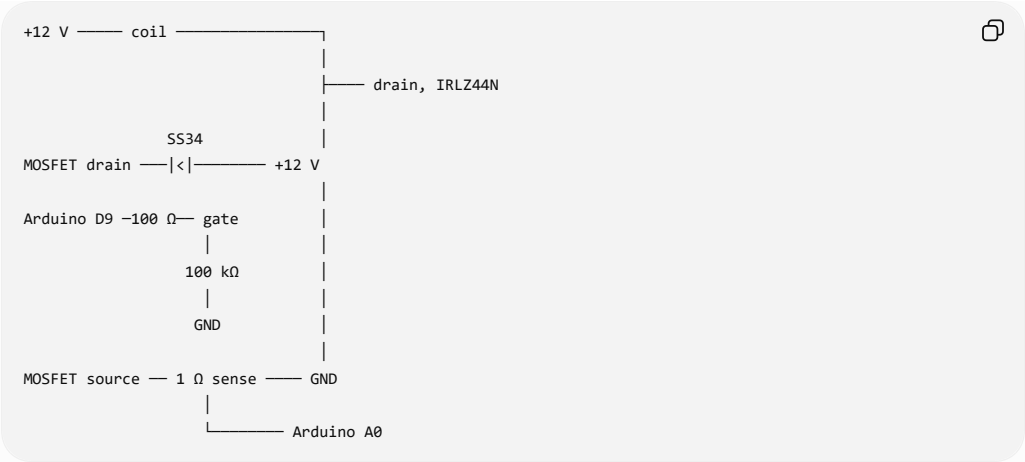
You must measure the actual resistance and inductance after winding.

A rough center-field estimate is:

$$B \approx \mu_0 \times N \times I / (2 \times r)$$

That formula is only an approximation. Measure the actual field using a calibrated Hall probe or gaussmeter.

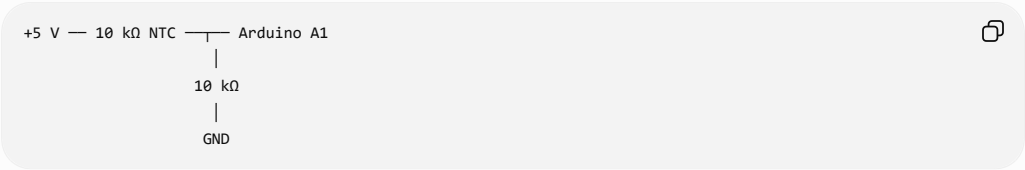
Circuit



The diode orientation is:

Anode → MOSFET drain
Cathode → +12 V

Temperature sensing:



Firmware behavior

The included Arduino program supports:

START
STOP
CLEAR
STATUS
FREQ 10
PULSE 500

It automatically stops when:

Current > 0.30 A
Coil temperature > 40 °C
Temperature sensor fails
Session reaches 60 seconds
Frequency or pulse width exceeds limits
Duty cycle exceeds 10%

The code outputs serial telemetry such as:

PULSE,time_ms=1250,frequency_hz=10,pulse_us=500,current_a=0.184,temperature_c=27.6

Arduino documents `delayMicroseconds()` for short pulse timing; for a later precision version, replace software delay timing with a hardware timer and verify the waveform using an oscilloscope. [Arduino Docum...](#)

Package contents

pemf_bench_v01/
├─ README.md
├─ firmware/
│ └─ pemf_bench_v01.ino
├─ hardware/
│ └─ schematic.md
├─ bom.csv
└─ coil_spec.md

```
└─ simulation/
   └─ coil_pulse.cir
```

Test it first using a **47 Ω power resistor instead of the coil**, then a measured R/L load, and finally the coil with the supply initially limited to 0.10 A. Verify the gate, drain, current waveform, flyback behavior, and temperature shutdown with proper instruments.

Do not apply this prototype to your daughter or anyone else. It is an engineering field-generation demonstrator, not a medical treatment device. `</>`

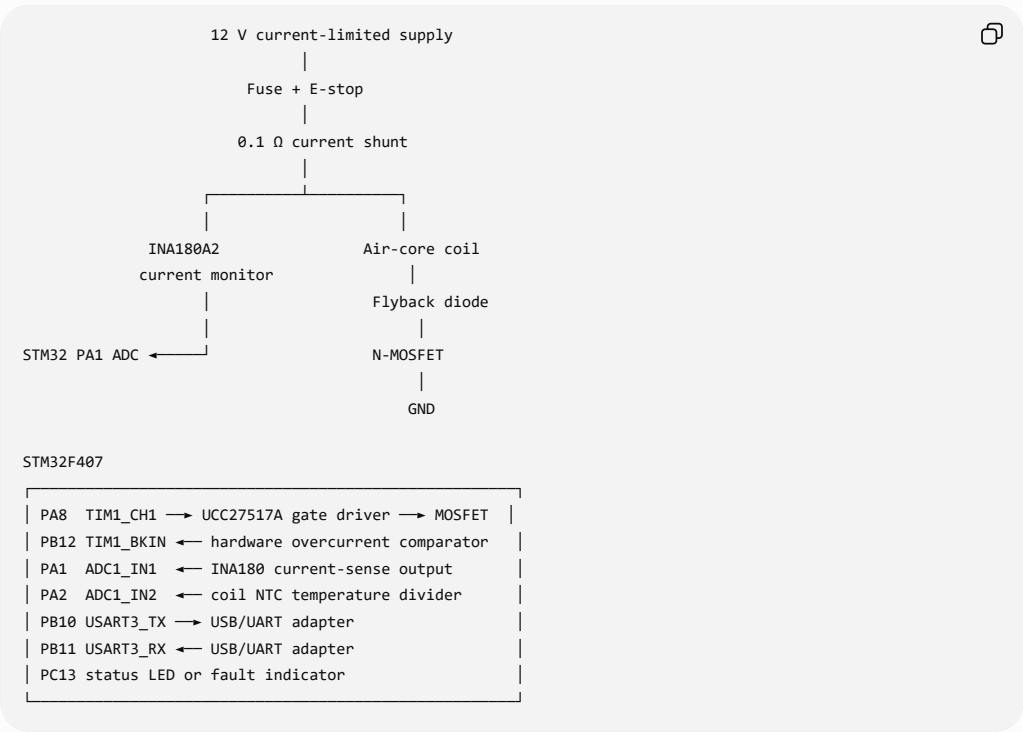
 Sources

I am going to build it STM32F407

Thought for 1m 20s >

STM32F407 is a solid choice for the **bench PEMF prototype**. It gives you hardware PWM, ADC with DMA, and—most importantly—the advanced timer’s **hardware break input**, so an overcurrent condition can shut off the coil without waiting for firmware. The STM32F407 runs at 3.3 V I/O levels and includes advanced PWM timers and multiple ADCs.  +1

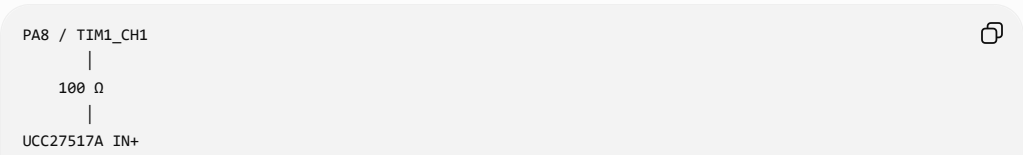
Updated STM32F407 architecture

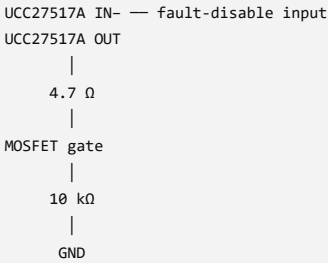


Critical change from the Arduino design

Do **not** drive the IRLZ44N directly from the STM32 GPIO.

The STM32 output is approximately 3.3 V, while an IRLZ44N-style power MOSFET is better driven with a higher gate voltage. Add a proper gate driver such as the **UCC27517A**, powered from approximately 10–12 V. It accepts 3.3 V control logic and has undervoltage protection that keeps the output low during invalid supply conditions. 





Recommended current sensing

Replace the previous 1 Ω shunt with:

Shunt: 0.1 Ω, 1%, 1 W
Amplifier: INA180A2
Gain: 50 V/V



At 0.30 A:

Shunt voltage = 0.30 A × 0.1 Ω
= 0.030 V

ADC voltage = 0.030 V × 50
= 1.50 V



That gives the STM32 ADC a much better signal while causing only a small voltage drop. The INA180 supports fixed gains including 50 V/V and can sense shunts at common-mode voltages up to 26 V.

Texas Instruments +1

Suggested pin assignment

This example works well for an STM32F407VG-class device:

Function	STM32 pin	Peripheral
Coil PWM	PA8	TIM1_CH1
Hardware fault shutdown	PB12	TIM1_BKIN
Current measurement	PA1	ADC1_IN1
Coil temperature	PA2	ADC1_IN2
Serial TX	PB10	USART3_TX
Serial RX	PB11	USART3_RX
Status LED	PC13 or board-specific LED	GPIO

PA8 can provide TIM1 PWM, and PB12 supports the TIM1 break input. STMicroelectron... +1

STM32CubeMX configuration

Clock

Example configuration:

System clock: 168 MHz
APB1: 42 MHz
APB2: 84 MHz
TIM1 clock: 168 MHz



TIM1 PWM

Use a 100 kHz timer counter:

```

Prescaler = 1679
Timer tick = 168 MHz / 1680
            = 100 kHz

```



For 10 Hz output:

```

Period = 100,000 / 10 - 1
        = 9,999

```



For a 500 µs pulse:

```

Pulse ticks = 500 µs × 100 kHz
              = 50 ticks

```



CubeMX settings:

```

TIM1:
  Clock Source: Internal Clock
  Channel 1: PWM Generation CH1
  Prescaler: 1679
  Counter Period: 9999
  Pulse: 50
  Counter Mode: Up
  Auto-reload preload: Enable

Break and Dead-Time:
  Break State: Enable
  Break Polarity: High
  Automatic Output: Disable
  Dead Time: 0

```



ADC1

```

ADC1:
  Resolution: 12 bit
  Scan conversion: Enable
  Continuous conversion: Enable
  DMA continuous request: Enable

Rank 1: ADC1_IN1 - current
Rank 2: ADC1_IN2 - temperature

```



USART3

```

Baud rate: 115200
Data bits: 8
Stop bits: 1
Parity: None

```



STM32 HAL control code

Add this to the CubeIDE-generated application.

```

#include "main.h"
#include <math.h>
#include <stdbool.h>
#include <stdio.h>

extern TIM_HandleTypeDef htim1;
extern ADC_HandleTypeDef hadc1;
extern UART_HandleTypeDef huart3;

#define PWM_TIMER_TICK_HZ    100000UL

#define MIN_FREQUENCY_HZ    1UL

```

```

#define MAX_FREQUENCY_HZ      50UL

#define MIN_PULSE_US          50UL
#define MAX_PULSE_US          1000UL

#define MAX_DUTY_PERCENT       10UL
#define MAX_SESSION_MS        60000UL

#define ADC_REFERENCE_V        3.3f
#define ADC_MAX_VALUE          4095.0f

#define SHUNT_RESISTANCE_OHM    0.1f
#define CURRENT_GAIN           50.0f

#define MAX_CURRENT_A           0.30f
#define MAX_COIL_TEMP_C         40.0f

static volatile uint16_t adcValues[2];

static bool pemfRunning = false;
static bool pemfFaulted = false;
static uint32_t sessionStartMs = 0;

</> C

```



Configure frequency and pulse width

```

static bool PEMF_SetParameters(
    uint32_t frequencyHz,
    uint32_t pulseUs)
{
    if (frequencyHz < MIN_FREQUENCY_HZ ||
        frequencyHz > MAX_FREQUENCY_HZ) {
        return false;
    }

    if (pulseUs < MIN_PULSE_US ||
        pulseUs > MAX_PULSE_US) {
        return false;
    }

    const uint32_t periodTicks =
        PWM_TIMER_TICK_HZ / frequencyHz;

    const uint32_t pulseTicks =
        (pulseUs * PWM_TIMER_TICK_HZ) / 1000000UL;

    if (periodTicks == 0 || pulseTicks == 0) {
        return false;
    }

    if ((pulseTicks * 100UL) >
        (periodTicks * MAX_DUTY_PERCENT)) {
        return false;
    }

    __HAL_TIM_DISABLE(&htim1);

    __HAL_TIM_SET_AUTORELOAD(
        &htim1,
        periodTicks - 1UL);

    __HAL_TIM_SET_COMPARE(
        &htim1,
        TIM_CHANNEL_1,
        pulseTicks);

    __HAL_TIM_SET_COUNTER(&htim1, 0UL);
}

```

```

configuredFrequencyHz = frequencyHz;
configuredPulseUs = pulseUs;

__HAL_TIM_ENABLE(&htim1);

```



Start and stop

```

static bool PEMF_Start(void)
{
    if (pemfFaulted) {
        return false;
    }

    if (!PEMF_SetParameters(
        configuredFrequencyHz,
        configuredPulseUs)) {
        return false;
    }

    if (HAL_TIM_PWM_Start(
        &htim1,
        TIM_CHANNEL_1) != HAL_OK) {
        return false;
    }

    sessionStartMs = HAL_GetTick();
    pemfRunning = true;

    return true;
}

static void PEMF_Stop(void)
{
    HAL_TIM_PWM_Stop(
        &htim1,
        TIM_CHANNEL_1);
}

```



Current calculation

```

static float PEMF_GetCurrentA(void)
{
    const float adcVoltage =
        ((float)adcValues[0] * ADC_REFERENCE_V) /
        ADC_MAX_VALUE;

    return adcVoltage /
}

```



NTC temperature calculation

For a 10 kΩ NTC with a 10 kΩ divider:

```

static float PEMF_GetTemperatureC(void)
{
    const uint16_t raw = adcValues[1];

    if (raw == 0 || raw >= 4095) {

```



```

        return NAN;
    }

    const float voltage =
        ((float)raw * ADC_REFERENCE_V) /
        ADC_MAX_VALUE;

    const float fixedResistance = 10000.0f;
    const float nominalResistance = 10000.0f;
    const float beta = 3950.0f;
    const float nominalTemperatureK = 298.15f;

    const float ntcResistance =
        fixedResistance *
        voltage /
        (ADC_REFERENCE_V - voltage);

    const float inverseTemperature =
        (1.0f / nominalTemperatureK) +
        (1.0f / beta) *
        logf(ntcResistance / nominalResistance);

```

</> C



Fault shutdown

```

static void PEMF_Trip(const char *reason)
{
    PEMF_Stop();
    pemfFaulted = true;

    char message[96];

    const int length = snprintf(
        message,
        sizeof(message),
        "TRIP,%s\r\n",
        reason);

    HAL_UART_Transmit(
        &huart3,
        (uint8_t *)message,
        (uint16_t)length,

```

</> C



TIM1 hardware-break callback

```

void HAL_TIMEx_BreakCallback(
    TIM_HandleTypeDef *htim)
{
    if (htim->Instance == TIM1) {
        PEMF_Trip("hardware_break");
    }
}

```

</> C



The PWM output is already disabled by TIM1 hardware before the callback executes. The callback records and latches the fault.

Main-loop supervision

```

static void PEMF_Process(void)
{

```

```

    if (!pemfRunning) {
        return;
    }

    const float currentA =
        PEMF_GetCurrentA();

    const float temperatureC =
        PEMF_GetTemperatureC();

    if (currentA > MAX_CURRENT_A) {
        PEMF_Trip("software_overcurrent");
        return;
    }

    if (!isfinite(temperatureC)) {
        PEMF_Trip("temperature_sensor_fault");
        return;
    }

    if (temperatureC > MAX_COIL_TEMP_C) {
        PEMF_Trip("overtemperature");
        return;
    }

    if ((HAL_GetTick() - sessionStartMs) >
        MAX_SESSION_MS) {
        PEMF_Stop();
    }

```

</> C



Initialization

```

int main(void)
{
    HAL_Init();
    SystemClock_Config();

    MX_GPIO_Init();
    MX_DMA_Init();
    MX_ADC1_Init();
    MX_USART3_UART_Init();
    MX_TIM1_Init();

    HAL_ADC_Start_DMA(
        &hadc1,
        (uint32_t *)adcValues,
        2);

    PEMF_SetParameters(10, 500);

    while (1) {
        PEMF_Process();
        HAL_Delay(5);
    }
}

```

</> C



Hardware shutdown path

Do not depend only on `PEMF_Process()`.

Use a comparator:

```

INA180 output
|
├── STM32 ADC
|
├── comparator
|

```



```
└─ PB12 / TIM1_BKIN
└─ gate-driver disable input
```

The emergency-stop button should also disable the gate driver or disconnect the 12 V coil supply directly:

```
E-stop
└─ opens coil supply
└─ asserts driver disable
```

This ensures that firmware crashes, debugger pauses, or ADC failures cannot leave the MOSFET continuously energized.

Recommended STM32 version of the power stage

```
12 V
|
└─ fuse
|
└─ E-stop
|
└─ 0.1 Ω high-side shunt
|
└─ coil
|
└─ MOSFET drain
MOSFET source
|
└─ GND
```

The INA180 measures across the high-side shunt, while the MOSFET remains a conventional low-side switch.

Use this only as a **bench magnetic-field generator** with test loads and a gaussmeter. It is not a human-use PEMF treatment design and should not be applied to your daughter or anyone else.

...

🚫🔪

Sources

give me the logical flow of the PEMF device

Thought for a couple of seconds >

Below is the logical flow for the STM32F407 PEMF bench device.

1. Overall signal flow

```
User command
|
▼
STM32 validates settings
|
▼
Safety checks
|
▼
TIM1 generates pulse waveform
|
▼
Gate driver amplifies 3.3 V control signal
|
▼
MOSFET switches coil current
|
▼
Coil produces pulsed magnetic field
|
▼
```

Current and temperature are measured
 |
 ▼
 STM32 continues, stops, or enters fault state

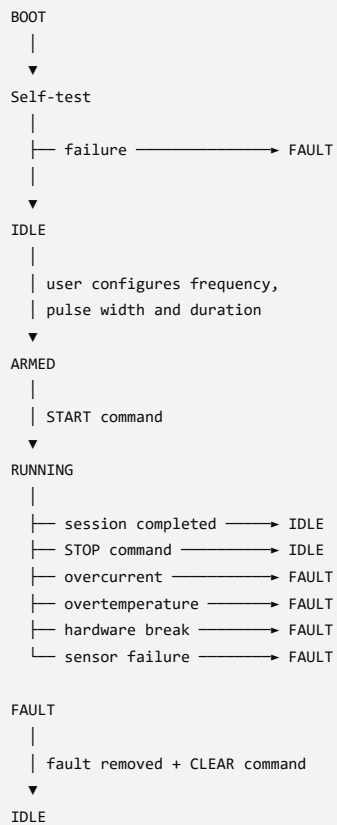
2. Device state machine

The software should use five main states:

```
typedef enum
{
    PEMF_STATE_BOOT,
    PEMF_STATE_IDLE,
    PEMF_STATE_ARMED,
    PEMF_STATE_RUNNING,
}
```

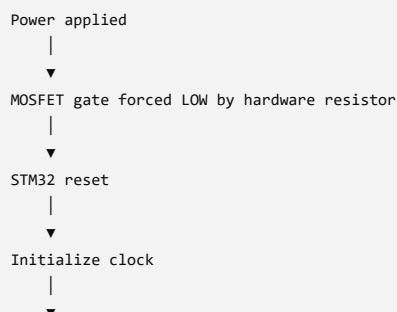
⌕ C

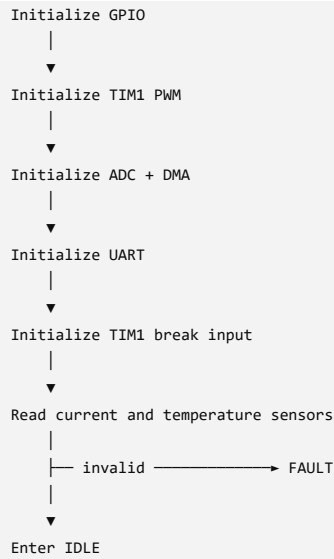
State flow:



3. Boot flow

At power-up:





The MOSFET output must remain disabled during the entire boot process.

4. Configuration flow

In the idle state, the user can set:

Frequency
Pulse width
Session duration
Maximum current
Maximum temperature

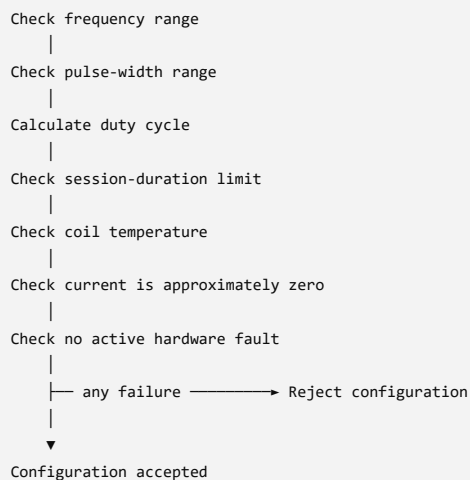


Example command:

FREQ 10
PULSE 500
DURATION 30
START



Before accepting the settings:



Example validation:

```

bool PEMF_ValidateParameters(
    uint32_t frequencyHz,
    uint32_t pulseUs,
    uint32_t durationMs)
  
```

```

{
    if (frequencyHz < 1 || frequencyHz > 50)
        return false;

    if (pulseUs < 50 || pulseUs > 1000)
        return false;

    if (durationMs == 0 || durationMs > 60000)
        return false;

    uint32_t periodUs = 1000000UL / frequencyHz;

    if (pulseUs >= periodUs)
        return false;

    float dutyCycle =
        ((float)pulseUs / (float)periodUs) * 100.0f;

    if (dutyCycle > 10.0f)
        return false;
}

```

⌞ C



5. Start sequence

When the user sends **START**:

```

START command
|
▼
Verify state is IDLE or ARMED
|
▼
Verify no latched fault
|
▼
Read current sensor
|
▼
Read temperature sensor
|
▼
Verify emergency stop is released
|
▼
Verify gate driver is ready
|
▼
Load TIM1 period
|
▼
Load TIM1 pulse width
|
▼
Reset timer counter
|
▼
Clear TIM1 break flags
|
▼
Enable PWM output
|
▼
Store session start time
|
▼
Enter RUNNING

```



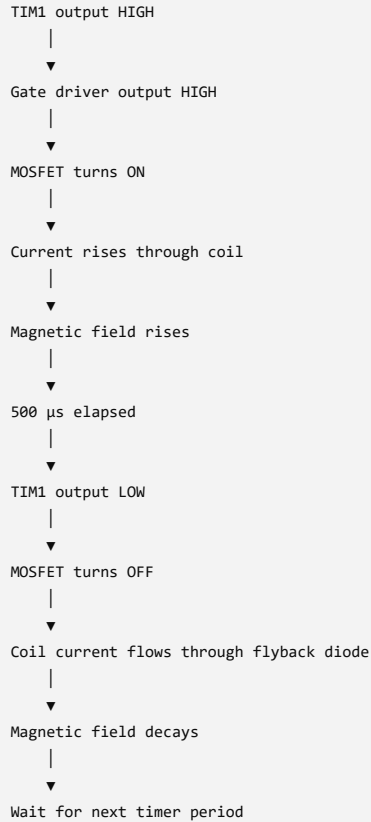
6. Pulse-generation flow

TIM1 produces the pulse waveform automatically.

For a 10 Hz, 500 μ s pulse:

Period = 100 ms
Pulse ON = 500 μ s
Pulse OFF = 99.5 ms

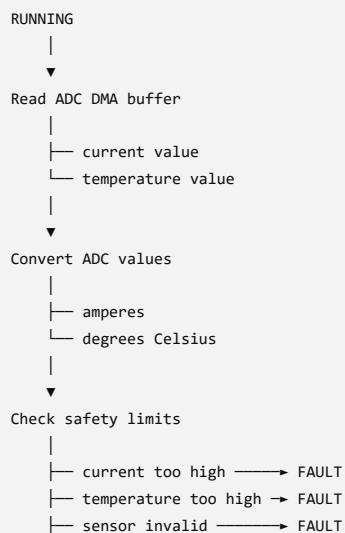
Electrical sequence:

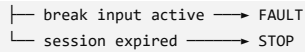


The STM32 does not need to manually toggle the pin for every pulse. Hardware TIM1 performs this more accurately.

7. Monitoring flow

While the device is running, the main loop continuously supervises the system.





Example task:

```

void PEMF_Process(void)
{
    switch (pemfState)
    {
        case PEMF_STATE_IDLE:
            break;

        case PEMF_STATE_ARMED:
            break;

        case PEMF_STATE_RUNNING:
            PEMF_MonitorCurrent();
            PEMF_MonitorTemperature();
            PEMF_CheckSessionTime();
            PEMF_SendTelemetry();
            break;

        case PEMF_STATE_FAULT:
            PEMF_ForceOutputOff();
            break;

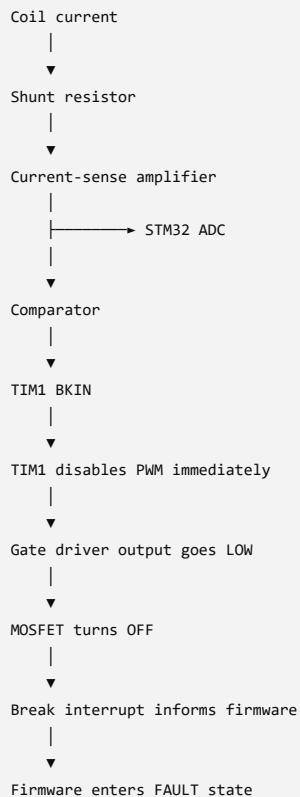
        default:
            PEMF_EnterFault(PEMF_FAULT_INVALID_STATE);
            break;
    }
}
  
```

⌕ C



8. Hardware overcurrent flow

The fastest protection path should not depend on firmware.



The hardware shutdown should happen before the interrupt callback.


```
void HAL_TIMEx_BreakCallback(TIM_HandleTypeDef *htim)
{
    if (htim->Instance == TIM1)
    {
        pemfFault = PEMF_FAULT_HARDWARE_OVERCURRENT;
        pemfState = PEMF_STATE_FAULT;
    }
}
```



9. Fault flow

Any serious fault causes a latched shutdown.

```

Fault detected
|
▼
Disable TIM1 output
|
▼
Disable gate driver
|
▼
Force MOSFET gate LOW
|
▼
Record fault code
|
▼
Turn fault LED ON
|
▼
Send fault message through UART
|
▼
Enter FAULT state
  
```



Example fault codes:

```
typedef enum
{
    PEMF_FAULT_NONE,
    PEMF_FAULT_OVERCURRENT,
    PEMF_FAULT_OVERTEMPERATURE,
    PEMF_FAULT_TEMP_SENSOR,
    PEMF_FAULT_HARDWARE_BREAK,
    PEMF_FAULT_EMERGENCY_STOP,
    PEMF_FAULT_DRIVER_NOT_READY,
}
```



A fault should remain latched until:

```

Output is stopped
Fault source is removed
Temperature is below reset threshold
Current is zero
User sends CLEAR
  
```



Do not automatically restart after a fault.

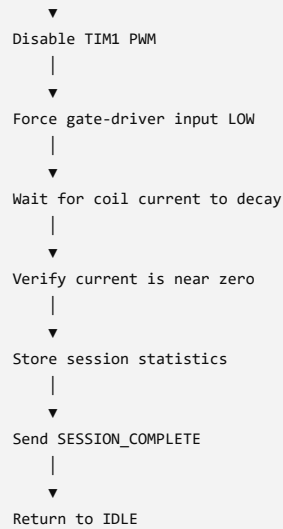
10. Normal stop flow

Normal session completion:

```

Session timer expires
|
  
```

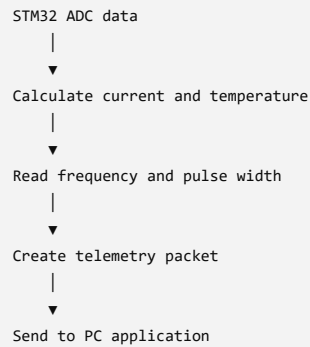




Manual stop follows the same sequence.

11. Telemetry flow

The STM32 can send data over UART or USB:



Example:

```

STATE=RUNNING,
TIME_MS=12450,
FREQ_HZ=10,
PULSE_US=5000,
CURRENT_A=0.182,
TEMP_C=29.4,
FAULT=NONE
  
```

A structured packet can be:

```

typedef struct
{
    uint32_t timestampMs;
    uint32_t frequencyHz;
    uint32_t pulseWidthUs;
    float currentA;
    float temperatureC;
    PEMF_State state;
}
  
```

12. Recommended firmware task structure

```

main.c
├── system initialization
  
```

```

└─ main state-machine loop

pemf_control.c
├─ parameter validation
├─ start
├─ stop
└─ state management

pemf_pwm.c
├─ TIM1 configuration
├─ frequency control
└─ pulse-width control

pemf_safety.c
├─ overcurrent
├─ overtemperature
├─ break input
├─ emergency stop
└─ fault latch

pemf_measurement.c
├─ ADC DMA
├─ current conversion
└─ NTC conversion

pemf_protocol.c
├─ UART commands
└─ telemetry output

```

13. Main-loop example

```

int main(void)
{
    HAL_Init();
    SystemClock_Config();

    MX_GPIO_Init();
    MX_DMA_Init();
    MX_ADC1_Init();
    MX_TIM1_Init();
    MX_USART3_UART_Init();

    PEMF_Init();

    while (1)
    {
        PEMF_ReadCommands();
        PEMF_ProcessStateMachine();
        PEMF_ProcessMeasurements();
        PEMF_ProcessSafety();
        PEMF_ProcessTelemetry();

        HAL_Delay(1);
    }
}

```

↔ C



The essential logic is:

```

Configure
↓
Validate
↓
Arm
↓
Generate pulses
↓
Measure current and temperature
↓
Continue while safe

```



↓
Stop normally or latch a fault

This flow is suitable for a **bench magnetic-field generator**, not a human-treatment device.



how to simulate the STM32F chip program working?

Thought for 41s >

The cleanest way to simulate your STM32F407 PEMF firmware is to use **three separate simulation layers**:

Firmware logic and STM32 peripherals → Renode
Coil, MOSFET and flyback circuit → SPICE
Final timing and safety verification → Real STM32F407 board



STM32CubeIDE builds and debugs the project, but it is not a complete electrical simulator for the MCU, MOSFET, coil, current sensor and magnetic field. Renode officially supports an STM32F4 Discovery platform and can load your compiled ELF firmware, expose UART output and connect to GDB. [Renode +2](#)

1. Prepare a simulation build

Create two CubeIDE configurations:

Debug_HW → real STM32F407 board
Debug_SIM → Renode simulation



For `Debug_SIM`, add this compiler definition:

PEMF_SIMULATION



In STM32CubeIDE:

Project
→ Properties
→ C/C++ Build
→ Settings
→ MCU GCC Compiler
→ Preprocessor
→ Defined symbols
→ PEMF_SIMULATION



2. Separate device logic from hardware access

Do not let the state machine read ADC registers directly.

Use an interface:

```
#ifndef PEMF_IO_H
#define PEMF_IO_H

#include <stdbool.h>

float PEMF_IO_GetCurrentA(void);
float PEMF_IO_GetTemperatureC(void);

bool PEMF_IO_IsEmergencyStopActive(void);
bool PEMF_IO_IsDriverReady(void);

void PEMF_IO_EnableOutput(void);
void PEMF_IO_DisableOutput(void);
```

C



Your main control code should only call these functions:

```
void PEMF_ProcessSafety(void)
{
    if (PEMF_IO_IsEmergencyStopActive())
    {
        PEMF_EnterFault(PEMF_FAULT_EMERGENCY_STOP);
        return;
    }

    if (!PEMF_IO_IsDriverReady())
    {
        PEMF_EnterFault(PEMF_FAULT_DRIVER_NOT_READY);
        return;
    }

    if (PEMF_IO_GetCurrentA() > 0.30f)
    {
        PEMF_EnterFault(PEMF_FAULT_OVERCURRENT);
        return;
    }

    if (PEMF_IO_GetTemperatureC() > 40.0f)
    {
        PEMF_EnterFault(PEMF_FAULT_OVERTEMPERATURE);
        return;
    }
}
```

C



3. Provide simulated sensor values

The current Renode STM32F4 platform models timers, GPIO, UART, DMA and several other peripherals, but its published STM32F4 platform description does not currently include an ADC model. Therefore, the easiest approach is to replace ADC measurements with controllable software values in the simulation build. [GitHub](#)

Create `pemf_io_sim.c`:

```
#include "pemf_io.h"

static volatile float simulatedCurrentA = 0.0f;
static volatile float simulatedTemperatureC = 25.0f;

static volatile bool simulatedEmergencyStop = false;
static volatile bool simulatedDriverReady = true;
static volatile bool simulatedOutputEnabled = false;

float PEMF_IO_GetCurrentA(void)
{
    return simulatedCurrentA;
}

float PEMF_IO_GetTemperatureC(void)
{
    return simulatedTemperatureC;
}

bool PEMF_IO_IsEmergencyStopActive(void)
{
    return simulatedEmergencyStop;
}

bool PEMF_IO_IsDriverReady(void)
{
    return simulatedDriverReady;
}
```

```

}

void PEMF_IO_EnableOutput(void)
{
    simulatedOutputEnabled = true;
}

void PEMF_IO_DisableOutput(void)
{
    simulatedOutputEnabled = false;
}

void PEMF_SIM_SetCurrent(float currentA)
{
    simulatedCurrentA = currentA;
}

void PEMF_SIM_SetTemperature(float temperatureC)
{
    simulatedTemperatureC = temperatureC;
}

void PEMF_SIM_SetEmergencyStop(bool active)
{
    simulatedEmergencyStop = active;
}

void PEMF_SIM_SetDriverReady(bool ready)
{
    simulatedDriverReady = ready;
}

bool PEMF_SIM_IsOutputEnabled(void)
{
    </> C

```



For the real board, compile `pemf_io_stm32.c` instead:

```

#include "pemf_io.h"
#include "main.h"

extern TIM_HandleTypeDef htim1;
extern volatile uint16_t adcValues[2];

float PEMF_IO_GetCurrentA(void)
{
    const float adcVoltage =
        ((float)adcValues[0] * 3.3f) / 4095.0f;

    return adcVoltage / (0.1f * 50.0f);
}

float PEMF_IO_GetTemperatureC(void)
{
    /* Real NTC calculation */
    return PEMF_ConvertNtcToTemperature(adcValues[1]);
}

bool PEMF_IO_IsEmergencyStopActive(void)
{
    return HAL_GPIO_ReadPin(
        ESTOP_GPIO_Port,
        ESTOP_Pin) == GPIO_PIN_RESET;
}

bool PEMF_IO_IsDriverReady(void)
{
    return HAL_GPIO_ReadPin(
        DRIVER_READY_GPIO_Port,

```

```
        DRIVER_READY_Pin) == GPIO_PIN_SET;
    }

    void PEMF_IO_EnableOutput(void)
    {
        HAL_TIM_PWM_Start(&htim1, TIM_CHANNEL_1);
    }

    void PEMF_IO_DisableOutput(void)
    {
        HAL_TIM_PWM_Stop(&htim1, TIM_CHANNEL_1);

        HAL_GPIO_WritePin(
            DRIVER_ENABLE_GPIO_Port,
            DRIVER_READY_Pin,
            GPIO_PIN_RESET);
    }
}
```

⌄ C



Select the implementation:

```
#ifdef PEMF_SIMULATION
    /* Build pemf_io_sim.c */
#else
    /* Build pemf_io_hw.c */
#endif
```

⌄ C



4. Add simulation commands

Use UART commands to inject conditions while the firmware is running:

```
SIM CURRENT 0.18
SIM TEMP 28
SIM ESTOP 0
SIM DRIVER 1

START
STATUS
```



Fault examples:

```
SIM CURRENT 0.35
```



Expected:

```
FAULT,OVERCURRENT
STATE=FAULT
OUTPUT=OFF
```



Temperature fault:

```
SIM CURRENT 0.18
SIM TEMP 45
```



Expected:

```
FAULT,OVERTEMPERATURE
STATE=FAULT
OUTPUT=OFF
```



Emergency-stop fault:

```
SIM TEMP 28
SIM ESTOP 1
```



Expected:

```
FAULT,EMERGENCY_STOP
STATE=FAULT
OUTPUT=OFF
```



Example command parser:

```
#ifdef PEMF_SIMULATION

#include <stdio.h>
#include <string.h>

void PEMF_SIM_ProcessCommand(const char *command)
{
    float value = 0.0f;
    int booleanValue = 0;

    if (sscanf(command, "SIM CURRENT %f", &value) == 1)
    {
        PEMF_SIM_SetCurrent(value);
        PEMF_SendMessage("OK,SIM_CURRENT\r\n");
        return;
    }

    if (sscanf(command, "SIM TEMP %f", &value) == 1)
    {
        PEMF_SIM_SetTemperature(value);
        PEMF_SendMessage("OK,SIM_TEMPERATURE\r\n");
        return;
    }

    if (sscanf(command, "SIM ESTOP %d", &booleanValue) == 1)
    {
        PEMF_SIM_SetEmergencyStop(booleanValue != 0);
        PEMF_SendMessage("OK,SIM_ESTOP\r\n");
        return;
    }

    if (sscanf(command, "SIM DRIVER %d", &booleanValue) == 1)
    {
        PEMF_SIM_SetDriverReady(booleanValue != 0);
        PEMF_SendMessage("OK,SIM_DRIVER\r\n");
        return;
    }

    PEMF_SendMessage("ERROR,UNKNOWN_SIM_COMMAND\r\n");
}

</> C
```



5. Build the ELF file

Build the simulation configuration:

```
Project
→ Build Configurations
→ Set Active
→ Debug_SIM
```



Then:

```
Project
→ Build Project
```



The resulting file will normally be:

```
YourProject/
└─ Debug_SIM/
```



 YourProject.elf

Keep the ELF file. Renode uses it directly and preserves symbols for debugging.

6. Install and run Renode

Renode provides a predefined STM32F4 Discovery platform. The supported board corresponds to an STM32F407-based Discovery development board. [🔗 Renode +1](#)

Create `pemf_f407.resc` :

```
using sysbus

$name?="PEMF_STM32F407"
$bin?=@C:/Projects/PEMF/Debug_SIM/PEMF.elf

mach create $name

machine LoadPlatformDescription
    @platforms/boards/stm32f4_discovery-kit.repl

cpu PerformanceInMips 125

showAnalyzer sysbus.usart3

macro reset
    """
        sysbus LoadELF $bin
    """

runMacro $reset

machine StartGdbServer 3333
```

Keep `machine LoadPlatformDescription` on one line in the actual file:

```
machine LoadPlatformDescription @platforms/boards/stm32f4_discovery-kit.repl
```

Run it:

```
<> PowerShell
```

Then, in the Renode Monitor:

```
start
```

The official Renode STM32F4 platform includes USART3 at the correct STM32F4 peripheral address, so your firmware's USART3 telemetry can appear in the UART analyzer. [🔗 GitHub](#)

7. Debug with GDB

Open another terminal:

```
<> PowerShell
```

Inside GDB:

```
target remote :3333
monitor start

<> gdb
```

Useful breakpoints:

```
break PEMF_Start
break PEMF_Stop
break PEMF_EnterFault
break PEMF_ProcessSafety
```

 `</> gdb`

Inspect state:

```
print pemfState
print pemfFault
print configuredFrequencyHz
print configuredPulseUs
```

 `</> gdb`

Change simulated values directly:

 `</> gdb`

Renode supports breakpoints, watchpoints, stepping, memory inspection and source-level debugging through its GDB server. Virtual time stops while the emulated CPU is halted, which is useful when debugging timers and state transitions. [RS Renode](#)

8. Simulate TIM1 output

Your firmware uses:

PA8
TIM1_CH1



The official Renode STM32F4 platform connects TIM1 channel 1 to PA8, so it can execute the timer and GPIO behavior used by your PWM code. [GitHub](#)

In Renode, inspect the timer:

 `sysbus.timer1`

Log accesses:

 `sysbus LogPeripheralAccess timer1`

Inspect GPIO A:

 `sysbus.gpioPortA`

Log GPIO accesses:

 `sysbus LogPeripheralAccess gpioPortA`

You can also enable firmware-function logging:

 `cpu LogFunctionNames true "PEMF_"`

Renode provides peripheral-access logging and function-name tracing for investigating embedded drivers and firmware execution. [RS Renode](#)

9. Handle the TIM1 break input in simulation

For the simulation build, provide a software command:

SIM BREAK



Implementation:

```
void PEMF_SIM_TriggerHardwareBreak(void)
{
    PEMF_IO_DisableOutput();

    pemfFault =
        PEMF_FAULT_HARDWARE_BREAK;

    pemfState =
        PEMF_STATE_FAULT;

    PEMF_SendMessage(
</pre>

```

<> C



Use the exact HAL callback logic:

```
void PEMF_HandleBreakEvent(void)
{
    PEMF_IO_DisableOutput();

    pemfFault =
        PEMF_FAULT_HARDWARE_BREAK;

    pemfState =
        PEMF_STATE_FAULT;
}

#ifndef PEMF_SIMULATION

void HAL_TIMEx_BreakCallback(
    TIM_HandleTypeDef *htim)
{
    if (htim->Instance == TIM1)
    {
        PEMF_HandleBreakEvent();
    }
}

</pre>

```

<> C



This lets you test the firmware response, although it does not prove the real comparator-to-BKIN shutdown timing.

10. Test scenarios

Run these tests before using hardware.

Normal operation

SIM CURRENT 0.18
SIM TEMP 28
SIM ESTOP 0
SIM DRIVER 1

FREQ 10
PULSE 500
DURATION 10
START



Expected state flow:

IDLE
→ ARMED
→ RUNNING
→ IDLE



Invalid frequency

FREQ 100



Expected:

ERROR, FREQUENCY_RANGE



Invalid duty cycle

FREQ 50
PULSE 5000



Expected:

ERROR, DUTY_CYCLE



Overcurrent

START
SIM CURRENT 0.35



Expected:

RUNNING
→ FAULT_OVERCURRENT
→ output disabled



Overtemperature

SIM CURRENT 0.18
SIM TEMP 45
START



Expected:

FAULT_OVERTEMPERATURE



Emergency stop

SIM ESTOP 1



Expected:

FAULT_EMERGENCY_STOP



Driver failure

SIM DRIVER 0
START



Expected:

START rejected
FAULT_DRIVER_NOT_READY



Restart after fault

START



Expected:

ERROR, FAULT_LATCHED



Then:

SIM CURRENT 0
SIM TEMP 28
SIM ESTOP 0
SIM DRIVER 1
CLEAR
START



Expected:

FAULT
→ IDLE
→ RUNNING



11. Simulate the coil circuit separately

Use LTspice, ngspice or another SPICE program for:

12 V source
Coil resistance
Coil inductance
MOSFET
Gate driver
Flyback diode
Current-sense resistor



Example simplified SPICE model:

```
VPOWER VPLUS 0 12

VGATE GATE 0 PULSE(
+ 0
+ 10
+ 1m
+ 100n
+ 100n
+ 500u
+ 100m
+ )

RCOIL VPLUS COIL_TOP 55
LCOIL COIL_TOP DRAIN 35m
```

```
M1 DRAIN GATE 0 0 NMOS_MODEL

D1 DRAIN VPLUS DIODE MODEL

</> spice
```



Measure:

- MOSFET drain voltage
- Coil current rise
- Current-decay time
- Flyback-diode current
- MOSFET power
- Coil average power



Renode does not calculate the physical coil current, MOSFET switching loss or magnetic flux. That belongs in the electrical simulation.

12. What simulation can prove

Feature	Renode	SPICE	Real board
State machine	Yes	No	Yes
UART protocol	Yes	No	Yes
Parameter validation	Yes	No	Yes
Fault latching	Yes	No	Yes
TIM1 register configuration	Yes	No	Yes
Software response to break	Yes	No	Yes
Coil current waveform	No	Yes	Yes
MOSFET switching transient	No	Yes	Yes
ADC noise and accuracy	No	Partial	Yes
Actual BKIN shutdown delay	No	Partial	Yes
Magnetic field strength	No	Calculated estimate	Yes, with probe
EMI and thermal behavior	No	Partial	Yes

QEMU is less suitable here because its official STM32 documentation supports only a limited selection of STM32 boards and explicitly notes that many STM32 series remain unsupported; STM32F407 Discovery is not listed among its currently documented STM32 machines.  qemu.org

The practical verification path is:

1. Host unit tests
2. Renode firmware simulation
3. SPICE power-stage simulation
4. STM32F407 Discovery board without power stage
5. Resistive dummy load
6. Low-current coil test
7. Oscilloscope and current-probe verification



This verifies your firmware before exposing the MOSFET or coil to meaningful power.



 Sources

I am looking for the simulation tool for STM32 chips

Thought for 20s >

For your **STM32F407 PEMF project**, the best simulation tool depends on what you want to test.

Best choice: Proteus VSM

Use **Proteus VSM** when you want to draw the circuit and watch the STM32 firmware interact with components such as:

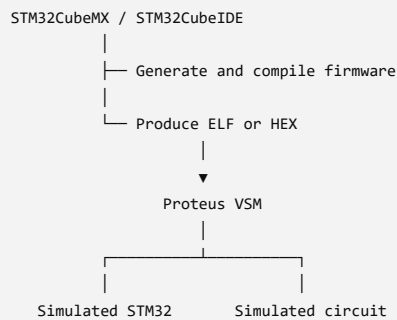
STM32F407
TIM1 PWM output
MOSFET gate driver
MOSFET
Coil / inductive load
Flyback diode
Current-sense amplifier
Comparator
UART terminal
Oscilloscope



Proteus combines microcontroller firmware simulation with schematic and SPICE-style circuit simulation. You can load the `.elf` or `.hex` generated by STM32CubeIDE, run it, use breakpoints, and observe virtual oscilloscopes and logic analyzers. [labcenter.com](#) +2

Before purchasing, verify that your exact STM32F407 package—such as STM32F407VGT6—is included in the selected Proteus VSM licence. Device coverage varies by processor package and licence module.

Recommended workflow:



Best free option: Renode

Use **Renode** when you mainly want to test:

- firmware state machines;
- UART commands;
- interrupts;
- timers;
- GPIO logic;
- fault handling;
- automated firmware tests.

Renode includes an STM32F4 Discovery example and can run the same ELF executable used on hardware. It also supports GDB debugging and virtual UART interaction. [Renode](#) +2

Renode does **not** realistically simulate the MOSFET, coil current, flyback diode, temperature, or magnetic field. Those sensor values normally need software models or injected test values.

Keil µVision Simulator

The **Keil µVision Simulator** is useful for CPU-level debugging and supported on-chip peripherals. It can simulate instructions, interrupts, memory, UARTs, and some peripheral timing without a physical board. Exact peripheral coverage depends on the selected MCU's simulation driver.  keil.com +2

It is useful for:

- C code execution
- Breakpoints
- Register inspection
- Interrupt tests
- UART simulation
- Logic validation



It is not the best choice for simulating your complete coil power circuit.

The MDK Community Edition is available free for qualifying non-commercial projects.  keil.arm.com

IAR C-SPY Simulator

IAR's C-SPY simulator can execute the target processor entirely in software and is useful for program-logic testing before hardware exists.  docs.iar.com +1

It is a professional commercial tool, but again, it is primarily a firmware debugger rather than a complete electrical-circuit simulator.

QEMU

QEMU currently includes STM32F405-based machine models, but its STM32 support remains partial. Its own documentation lists important missing peripherals including DMA, GPIO, I²C, DAC, CAN and others. That makes it a poor fit for an STM32F407 application using TIM1, ADC/DMA, GPIO and hardware safety inputs.



STM32CubeIDE itself

STM32CubeIDE is for:

- Project creation
- Code generation
- Compilation
- Programming
- Debugging on real hardware



It does not provide a complete virtual STM32 chip and circuit simulator. ST describes it as an editor, build environment and hardware-debugging tool.  STMicroelectron... +1

My recommendation

For your current design:

Proteus VSM

- STM32F407 firmware
- PWM
- MOSFET driver
- MOSFET
- coil
- flyback diode
- current sensing
- virtual oscilloscope



Then use:

- STM32CubeIDE → write and compile firmware
- Proteus VSM → simulate MCU and basic circuit
- LTspice → accurately analyze MOSFET and coil transients
- Real board → verify timing, protection, temperature and field



So the most suitable single tool for what you are describing is **Proteus VSM**. Renode is the strongest free choice for firmware-only simulation.

    ...  Sources

give me the download link for the Proteus VSM

Thought for 5s >

Official Proteus VSM download:

[Download the Proteus Design Suite free trial ↗](#)

Enter a valid email address, and Labcenter will send you a download link. The installer is about **600 MB**, and the microcontroller-simulation features are available for **14 days** in the trial. 

Official product pages:

- [Proteus VSM overview ↗](#)
- [Proteus pricing and licences ↗](#)
- [Proteus tutorials ↗](#)

Before purchasing, confirm that the licence includes your exact MCU, such as **STM32F407VGT6**, because processor support depends on the selected VSM package.

    ...  Sources

Proteus VSM is different from Proteus Professional 9.12?

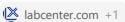
Thought for 12s >

They are **not completely separate programs**.

Proteus Design Suite is the complete software package. **Proteus VSM** is the microcontroller and circuit-simulation technology inside certain Proteus licences.

```
Proteus Design Suite
├─ Schematic Capture
├─ PCB Layout
├─ SPICE circuit simulation
├─ Proteus VSM
│  └─ Microcontroller simulation
│     └─ Firmware execution
│     └─ Virtual oscilloscope
│     └─ Logic analyser
│     └─ Peripheral models
└─ Optional processor-family licences
```



Labcenter describes Proteus VSM as the system-level environment that combines microcontroller simulation with schematic and electronic-circuit simulation.  +1

What “Proteus Professional” means

“Proteus Professional” usually refers to an **edition or licence bundle** of Proteus Design Suite. Depending on its licence, it might contain:

- schematic drawing only;
- PCB design;
- analog/digital SPICE simulation;
- VSM microcontroller simulation;

- support for only selected MCU families.

Therefore, having Proteus Professional installed does **not automatically guarantee STM32 simulation**.

Proteus VSM packages are sold according to processor families, and each VSM purchase includes schematic capture, the simulation engine, virtual instruments and peripheral models. [labcenter.com](#)

About “Proteus Professional 9.12”

I could not verify an official Labcenter release named **Proteus 9.12**. The currently advertised official release is **Proteus 9.2**, not 9.12. [labcenter.com](#) +1

So “Proteus Professional 9.12” may be:

- a typo for **9.2**;
- a modified or unofficial installer name;
- a reseller-specific package name;
- confusion with an older version such as **8.12**.

Be careful with unofficial downloads because they may contain modified executables or malware.

How to check your installation

Open Proteus and check:

Help → About Proteus



Confirm the actual version and licence information.

Then open:

Library → Pick Devices



Search for:

STM32F407VGT6
STM32F407VG
STM32F407



Finding a schematic symbol alone is not enough. Open the component properties and check whether it supports:

Program File
Clock Frequency
ELF or HEX firmware
VSM CPU model
Debug file



Try assigning your STM32CubeIDE output:

project.elf



When simulation starts successfully and the MCU executes firmware, your licence includes the necessary VSM support.

What you need for your project

For the STM32F407 PEMF bench circuit, your Proteus installation should include:

Proteus VSM
ARM Cortex-M / STM32 processor support
STM32F407 device model
Mixed-mode/SPICE simulation
Virtual oscilloscope
Logic analyser
UART terminal



The practical distinction is:

Name	Meaning
Proteus Design Suite	Entire software platform
Proteus Professional	An edition or licence bundle
Proteus VSM	MCU and system simulation engine
ARM/STM32 VSM package	Licence enabling supported STM32 simulation

Also note that the official trial allows microcontroller simulation for 14 days, but it restricts saving and simulating newly created custom designs; it is mainly intended for evaluating included examples.

 labcenter.com

    ...  Sources

so, can I still use the Proteus Professional for STM32 simulation?

Thought for 12s >

Yes—**Proteus Professional can simulate STM32**, but only when your licence includes the correct **Proteus VSM processor package**.

For **STM32F407**, you need:

Proteus Professional
+ Proteus VSM simulation engine
+ ARM Cortex-M4 processor package
+ STM32F407 device model



STM32F407 uses the **ARM Cortex-M4** core. Labcenter sells Cortex-M4 VSM support as a specific processor-family package; having only the Professional schematic or PCB edition is not enough.  labcenter.com +1

Check your installation

1. Search for the MCU

Open:

Library → Pick Devices



Search for:

STM32F407VGT6
STM32F407VG
STM32F407



2. Check whether it is simulatable

Place the component and open its properties. You should see fields such as:

Program File
Clock Frequency
Processor Clock



Load the firmware generated by STM32CubeIDE:

YourProject.elf



Proteus recommends ELF/DWARF when available because it preserves debugging information; HEX can also be used but normally provides less source-level debugging. [labcenter.com](#)

3. Start simulation

Connect something simple first:

STM32 PA8 / TIM1_CH1
↓
Logic analyzer or oscilloscope

Load the ELF file and press **Run**. When the STM32 executes the program and the virtual oscilloscope shows PWM, your installation supports the required VSM model.

Interpretation

What you find	Meaning
STM32F407 exists and firmware runs	Your licence supports it
STM32F407 exists but has no Program File field	It may be only a schematic symbol
"No simulation model" appears	Cortex-M4 VSM support is missing
STM32F407 is absent	Your version/library does not contain that model
Other STM32 models work but F407 does not	Exact-device support is missing

The decisive factor is **the installed VSM licence and device model, not the words "Proteus Professional."** Current Proteus releases also advertise STM32CubeIDE integration, but you still need the matching Cortex-M4 simulation package. [labcenter.com](#)

For your PEMF bench simulation, test the STM32 first with only TIM1, UART, LEDs, and virtual instruments. Add the MOSFET, diode, current sensing, and coil after confirming that the firmware runs correctly.

Sources

Yesterday 11:22 PM

write all the discussions into a md file

Done — I wrote the full discussion into a Markdown file.

[pemf_metatron_mora_zapper_full_discussion.md](#)

can you provide me the all the documents for metatron, mora, zapper equipments?

Worked for 1m 33s >

Yes. I prepared a **document catalog** for the public materials I could find for **Metatron/NLS, MORA, and Zapper equipment**.

[metatron_mora_zapper_document_catalog.md](#)

A few important notes:

I can provide **public manuals, official pages, product pages, and evidence-review documents**, but I cannot guarantee "all documents" because some materials are private, bundled with licensed software, behind

reseller portals, or protected by copyright/licence.

The catalog includes:

- **Metatron / NLS / Metapathia / Hunter 4025:** official IPP pages, software/etalon overview, device principle page, older Metapathia-GR Hunter manuals, 9D/4025 reseller references.
- **MORA / MORA Nova:** official Med-Tronik user manual, product page, network/Satellite setup document, evidence-review documents.
- **Zapper / Hulda Clark / frequency-generator devices:** SyncroZap manual, VariZapper manual, zapper instructions, schematic/reference pages, FTC warning/case materials.

Some key sources: IPP says HSS Metatron contains more than **12,000 reference etalons** and lists etalon groups such as organ models, microorganisms, allergens, homeopathy, food products, and others.

 The official Metatron device page describes the claimed trigger-sensor and computer-model workflow.  Med-Tronik’s MORA Nova manual states that MORA Nova is used for measuring skin electrical resistance with a weak current, and it also lists contraindications such as pregnancy, electronic implants, epilepsy, psychosis, organ transplant, and acute life-threatening disease situations. 

The VariZapper manual describes it as a bio-frequency generator and states it is not a medical device.



I did **not** find public engineering documents containing complete schematics, firmware, PCB files, raw databases, AI models, calibration procedures, or validated patient datasets for Metatron, MORA, or Zapper systems.

      Sources

give me the frequency info for each disease in Metatron, Mora, zapper

Worked for 1m 2s >

For **Metatron, MORA, and Zapper**, there is **no reliable public “frequency for each disease” table** that can be treated as medical or engineering truth.

The important point is this:

Device frequency range ≠ disease frequency
Vendor therapy program ≠ validated medical frequency
Etalon/similarity database ≠ open Hz table



1. Metatron / Metatron 9D / NLS

Metatron does **not** publicly provide disease frequencies in normal Hz form.

It works more like this:

Headset / trigger sensor
↓
Software scan
↓
Comparison with proprietary “etalon” database
↓
Similarity / Nidus / report output



The official Metatron software page says HSS Metatron contains **more than 12,000 reference etalons**, including groups for organ models, microorganisms, viruses, helminths, fungi, antibacterial drugs, antiviral drugs, antiparasitic drugs, and antifungal drugs. But it does **not** publish the numerical frequency arrays or disease-by-disease Hz values. 

So for Metatron:

Disease/category	Public frequency info
Bacteria	Proprietary etalon group; no public Hz table
Viruses	Proprietary etalon group; no public Hz table
Fungi	Proprietary etalon group; no public Hz table
Parasites / helminths	Proprietary etalon group; no public Hz table
Allergies	Proprietary etalon group; no public Hz table
Cancer/tumor claims	No public validated disease frequency
Growth/height	No valid frequency info

Conclusion: Metatron frequency information is locked inside the licensed software/database. Public documents describe the database categories, not the actual frequencies.

2. MORA / MORA Nova

MORA is different. It is a bioresonance feedback system. Public specifications describe the **device operating range**, not a reliable disease-frequency table.

A MORA Nova product page lists:

Frequency range:0.1 Hz to 1 MHz

Filter adjustment range:1 Hz to 400 kHz

Amplification:0.1 to 1,000,000 depending on channel/mode





Med-Tronik describes MORA Nova SRM as a device for measuring skin electrical resistance using a weak current via electrode and counter-electrode on skin measurement points. 

So for MORA:

Disease/category	Public frequency info
Allergy	Usually handled through programs, testing points, or ampoule/cup workflows; no public validated Hz table
Parasite	No official disease-specific Hz table found
Virus	No official disease-specific Hz table found
Bacteria	No official disease-specific Hz table found
Cancer	No official validated disease frequency
Pain/inflammation	Program-dependent; no validated universal Hz value
Growth/height	No valid frequency info

Conclusion: MORA has a public **technical frequency range**, but not a trustworthy public list like “TB = X Hz, cancer = Y Hz, allergy = Z Hz.”

3. Zapper

Zapper devices are simpler. The classic Hulda Clark style is usually a **positive-offset square-wave generator**.

Public Zapper sources commonly mention:

Classic Clark Zapper: about 30 kHz / 30,000 Hz

Some descriptions: 30–33 kHz

Some programmable devices: broad ranges such as 1 Hz to 150 kHz or more

A Clark-zapper FAQ describes the original famous Clark zapper as a **30 kHz** positive-offset square-wave generator. [clarkzapper.com](#) Another Clark-related source says Clark published the original **30 kHz** zapper and a lower **1 kHz** zappicator output. [huldaclark.com](#)

But the FTC has acted against marketers making unsubstantiated claims that Zappers kill bacteria, viruses, and parasites or treat serious diseases such as cancer, AIDS, diabetes, multiple sclerosis, Alzheimer’s, asthma, and other conditions. [FTC Federal Trade C...](#) +1

So for Zapper:

Disease/category	Public “frequency” info
General Clark-style use	Usually 30 kHz positive-offset square wave
Parasites	Often claimed with 30 kHz or programmable lists, but not validated
Bacteria	Claimed by sellers, not validated
Viruses	Claimed by sellers, not validated
Cancer	Claims specifically challenged by FTC
Diabetes/MS/Alzheimer’s/asthma	Claims specifically challenged by FTC
Growth/height	No valid frequency info

Conclusion: Zapper has the clearest public frequency number—**around 30 kHz**—but it is not disease-specific and not clinically validated.

Practical comparison

Device	Public frequency information	Disease-specific frequency table?
Metatron 9D/NLS	Proprietary etalon database; categories public	No
MORA Nova	Device range: about 0.1 Hz–1 MHz	No reliable public table
Zapper	Classic output around 30 kHz	No validated disease table

Best way to think about it

Metatron:
disease/category → proprietary etalon comparison
not public Hz

MORA:
disease/category → therapy/testing program
device can operate over wide frequency range
not public validated Hz

Zapper:
usually one or more generated square-wave frequencies
classic value ≈ 30 kHz
disease claims not validated

For your database design

For research, I would not store these as “medical disease frequencies.” Store them as **claimed vendor/program frequencies**:

```
condition_name
device_type
claimed_frequency_hz
waveform
source
evidence_level
validated_clinical_reference
notes
```

And set:

```
evidence_level = "unvalidated / vendor claim"
```

For actual health measurement, your **calibrated bioimpedance or PEMF bench system** is much better because it measures real physical values like ohms, phase, current, pulse width, magnetic field, and temperature.

📄 👍 💬 ↻ ⋮ 🌐 📄 Sources

write all the discussions in this channel into a md file

☰ 📁 SMS Search Done — I wrote the full channel discussion into a Markdown file.

🔗 ⬆ ⋮

📄 [full_channel_discussion_metatron_mora_zapper_pemf.md](#) </>

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Today 1:08 AM

give me the picture of the mora, zapper devices. give me url



Here are representative pictures and source URLs for **MORA** and **Zapper** devices.

MORA device pictures / URLs

MORA Nova / Med-Tronik type device

This is the desktop bioresonance unit with touchscreen, front ports, cups, and probe/electrode connectors.

Source pages: MedicalExpo product image/page and Med-Tronik devices page [🔗](#) Med-Tronik

MORA Nova product information

This page describes MORA Nova and its device/program functions. [🌐](#) Mora Turkey

Zapper device pictures / URLs

Hulda Clark-style Zapper with copper hand tubes

This is the common zapper form: a small controller box, wires, and two copper hand grips.

Dr. Clark / SyncroZap product pages

These show commercial Clark-style zappers with copper tubes or wristbands.  Dr. Hulda Clark +1

Dr. Clark Store zapper FAQ / accessory page

This page explains copper tubes, conductive wristbands, and connection accessories.  Dr. Clark Store

Generic 30 kHz Hulda Clark Zapper picture

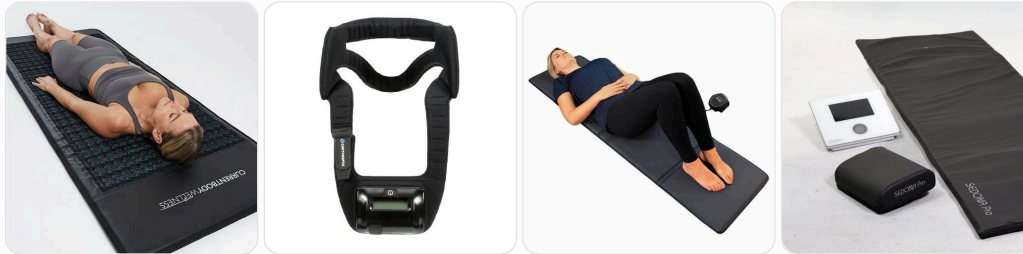
This shows another common simple 30 kHz zapper layout: small box, red/black wires, and copper handles.

The appearance varies a lot by reseller, especially for Zappers. MORA units are usually bigger desktop systems, while Zappers are usually handheld battery-powered frequency generators with copper tubes or wristbands.

    ...  Sources

give me the picture for the pemf devices

Here are representative PEMF device pictures. The main visual types are **full-body mats**, **portable/local applicators**, and **prescription-style bone-growth stimulators**.

**Source URLs****Full-body PEMF mat examples**

- CurrentBody Wellness Infrared PEMF Mat: <https://us.currentbody.com/products/currentbody-wellness-infrared-pemf-mat> ↗
- NewMed PEMF full-body mat/product category: <https://www.newmedltd.co.uk/> ↗
- Sedona Pro PEMF Mat: <https://www.mylongevitystudio.com/products/sedona-pro-complete-set-pemf-therapy-mat> ↗

Medical/bone-growth PEMF examples

- Orthofix Bone Growth Therapy: <https://bonegrowththerapy.com/> ↗
- FDA / AccessGUDID PhysioStim PEMF bone-growth stimulator listing: <https://accessgudid.nlm.nih.gov/devices/18257200131969> ↗

What you are seeing

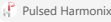
A **PEMF mat** is usually a flexible pad with embedded coils and a controller box. A **bone-growth PEMF stimulator** is usually a shaped applicator/brace with coils positioned around the fracture or fusion area. The mat-style products are mostly wellness-market devices, while bone-growth stimulators are medical devices for specific indications.

    ...

I want to see the antennas of pemf devices

Worked for 20s >

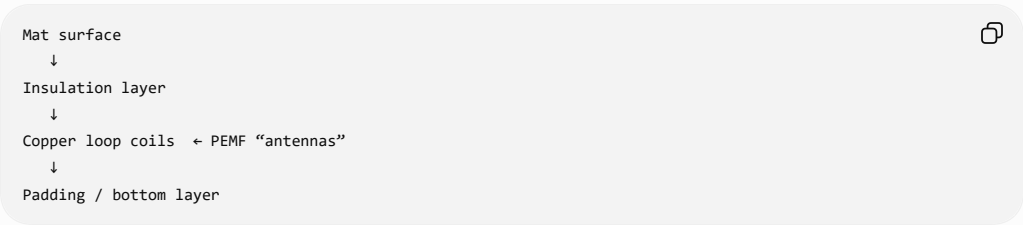


In PEMF devices, the “antenna” is usually **not** an RF antenna like Wi-Fi or radio. It is usually a **coil applicator**: copper wire loops that carry pulsed current and generate a pulsed magnetic field. PEMF product pages describe the applicator coil as the part that produces the magnetic field. 

What the PEMF “antenna” looks like

1. Full-body mat coil array

Inside a PEMF mat, you usually see several flat circular copper coils:



Some mat diagrams show multiple copper coils spread along the mat for body coverage.

2. Exposed large circular coils

Some opened mats show **large flat copper loops** embedded in foam. These are the actual field-generating parts, not the visible fabric cover.

3. Small local applicator coil

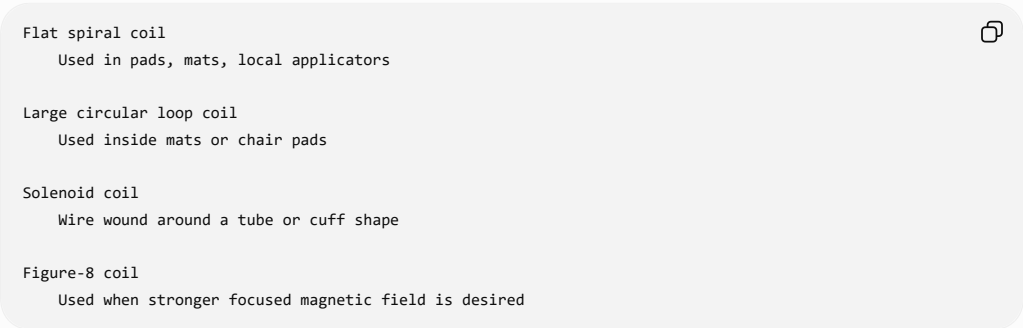
A smaller PEMF device may use one round or spiral coil for a knee, shoulder, hand, back, or local area:



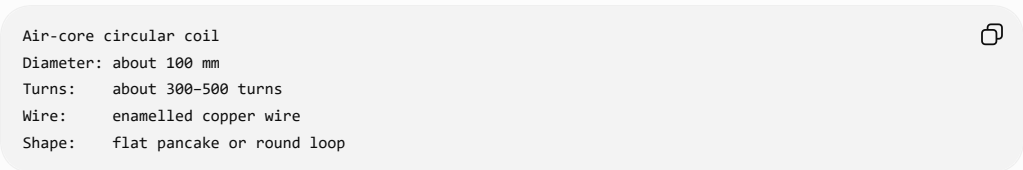
4. Bone-growth stimulator coil

Medical bone-growth PEMF stimulators often use a shaped coil applicator or brace-like applicator placed around the treatment area. It may not expose the copper coil, but internally it is still a coil structure.

Simple coil types used in PEMF



For your STM32F407 bench prototype, the easiest “antenna” is this one:



Like this conceptually:

circular copper
winding / coil



STM32 PWM → gate driver → MOSFET → coil → pulsed magnetic field

For design work, call it **coil**, **applicator coil**, or **PEMF coil** rather than antenna. That will help you find better technical drawings and product pictures.

    ...  Sources